Ramsey-Free Bounds for Two-Stage Shared-Decoder Composition

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Abstract

Section 6 of [6] studied two-stage compositions $K_2 \circ K_1$ of linear class-uniform shared decoders on $\mathrm{Mat}_{d\times d}(F^b)$ with M_1 and M_2 column classes, using van der Waerden and Hales–Jewett arithmetic progressions to bound d when $K_2 \circ K_1 = \tau$ and to show that $\mathrm{rank}_F(\tau - K_2 \circ K_1)/(bd) \to \infty$ as $d \to \infty$ at fixed M_1, M_2 when $(d : F) \neq 0$. No multitape Turing machine is known to induce such a two-stage linear class-uniform factorization; working within the same algebraic model, this paper replaces arithmetic progressions by Ramsey-free additive structures and proves two main composition bounds in Lean 4.

First (Theorem 3.3), over any field F, replacing progressions by affine Hilbert cubes extracted via iterated Cauchy–Schwarz proves that $K_2 \circ K_1 = \tau$ forces

$$d \;\le\; \max\bigl(8M_1M_2^2,\; (2M_1)^{2^{M_2}}\bigr).$$

This rules out exact two-stage linear factorizations for growing class schedules up to $M_1 = o\bigl(d^{1/2^{M_2}}\bigr)$ at fixed M_2 (including 3-AP-free Behrend partitions with $M_1 = 2^{O(\sqrt{\log d})}$) and $M_2 \le (1-\varepsilon)\log_2\log_2 d$ at fixed M_1, though at small constant class counts such as $M_1 = M_2 = 2$ it gives $d \le 256$ where [6, Corollary 6.4] gives $d \le 16$. Second (Theorem 4.6), over any field F with $(d : F) \neq 0$, combining shifted $(2M_2)$-fold monochromatic sumsets, arbitrary-subset Vandermonde pullbacks, sparse annihilating polynomials, and iterated Kővári–Sós–Turán extraction proves that for all $M_1, M_2 \ge 1$ and all $d \ge D_0(M_1, M_2)$,

$$\mathrm{rank}_F(\tau - K_2 \circ K_1) \;\ge\; c(M_1, M_2)\, b\, d\, (\log d)^{\frac{1}{2M_2-1}},$$

giving $\Omega(b\,d\log d)$ when $M_2 = 1$ (Corollary 4.8), with explicit constants where $D_0(1,1) = 16^{36} + 2 \approx 2.23 \times 10^{43}$. Finite-dimensional controls at $d = 13$, $d = 4$, and $d = 12$ test the hypotheses of both bounds.

Three shorter sections record companion results around [3]: (i) explicit three-term swap codes $\sum_{t=0}^{2} A_t X B_t^{\mathsf{T}} = X^{\mathsf{T}}$ over every field exceeding $d(d+1)/2$ at $d \in \{6, 7, 9, 12\}$ together with an all-field proof at $d = 4$, answering [3, Remark 5.6] at those dimensions (while $d = 5$ and every $d \ge 8$ outside $\{9, 12\}$ remain open), plus an observation on [3, Problem 9.3]; (ii) an Operator Schmidt Rank pass lower bound ($t \ge \log_2 d$) for a restricted pipeline of Stone's shuffle/unshuffle with row-local and column-local operators, paired with verified $O(L)$-time multitape Turing machines for shuffle, unshuffle, and within-row shear showing why $\mathrm{rk}_\otimes$ does not answer [3, Problem 9.5], alongside anti-diagonal tensor untwisting of shift-collision equations; and (iii) cut-causal cone evaluation on frozen `Transpose.TM` fibres, a Lean formalization of the $1/(\Delta+1)$ random-order bound of [4, Proposition 4.14(a)–(b)], and a negative observation explaining why column classes cannot absorb per-entry cut labels ([3, Problem 9.4(b)]). Question (T) remains open.

Contributions. Hans Brende chose the problem, fixed the verification standard, and directed the research. Gemini 4 Carbon (Google DeepMind) developed the mathematics, wrote the Lean 4 formalizations, and wrote the manuscript under his direction. Claude Fable 5.2 (Anthropic) conducted an external review of an earlier draft that identified vacuous statements in a preliminary sweep-extraction section, corrected cross-paper attributions, and recommended refocusing the paper around two-stage shared-decoder composition and moving the verified shuffle and shear machines to `lean-machine`.

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1 Introduction

1.1 Context and scope

Let $d \geq 2$, let $b = \lceil \log_2 d \rceil$ be the canonical entry bit-width, and let $L = d^2 b = \Theta(d^2 \log d)$ be the bit-length of a $d \times d$ matrix of b-bit entries laid out row by row on the read-only input tape of a deterministic multitape Turing machine with k read-write work tapes and a one-way write-only output tape ([5, Section 2.1 and Definition 2.1]). Question (T) asks whether every such machine that outputs the transposed matrix X^{T} for all inputs X requires $\omega(L)$ steps; [5, Conjecture 9.1] formulates a stronger asymmetric bipartite storage-coding conjecture (the Tape Coding Conjecture) that implies a negative answer to (T), and by Harvey and van der Hoeven [8, Corollary 6.9], if no linear-time multitape transposer exists then no linear-time multitape integer multiplier exists either.

The synthesis paper *Order, Symmetry and the Swap* [3] organized the algebraic obstacles studied across the series into two one-stage mechanisms on $\mathrm{Mat}_{d\times d}(F)$: an order bound for two-term operators $A_1 X B_1^{\mathsf{T}} + A_2 X B_2^{\mathsf{T}} = X^{\mathsf{T}}$ giving dimension at most $d(d+1)/2$ ([3, Theorem 5.1 and Corollary 5.5]), and a translate-pigeonhole bound (Theorem TP, [3, Definition 6.3 and Theorem 6.4]) for class-uniform rules in which column c of the output is computed by applying one of M shared linear decoders to the column-rotated matrix $\mathrm{rot}_c(X)_{i,j} = X_{i,j+c}$. As emphasized in [3, Sections 4.2 and 4.6], abstract storage-code models without shared rules admit high-rate network coding, whereas the class-uniform model captures translation covariance along a single regular sweep.

In the sequel *Capacity and Composition of Shared Transposition Decoders* [6, Section 6], Brende and GPT-6 Astra initiated the study of *two-stage compositions* $K_2 \circ K_1$ of linear class-uniform operators on $\mathrm{Mat}_{d\times d}(F^b)$, where K_1 has M_1 column classes and K_2 has M_2 column classes:

- For exact factorizations $K_2 \circ K_1 = \tau$ over any field F, [6, Theorem 6.3] reduced two-stage composition to a common-message subspace inside a cyclic channel module and extracted a monochromatic arithmetic progression in the stage-1 partition, proving $d \leq M_2(\mathsf{W}(M_1, M_2 + 1) - 1)$, where W denotes the van der Waerden number; at $M_1, M_2 \leq 2$, [6, Corollary 6.4] used $\mathsf{W}(2,3) \leq 9$ to obtain the small-parameter bound $d \leq 16$.

- For approximate factorizations over any field F with $(d : F) \neq 0$, [6, Theorem 6.6] proved that $r := \mathrm{rank}_F(\tau - K_2 \circ K_1)$ satisfies

$$r \;\geq\; \frac{bk(d - N_0 + 1)}{2}, \qquad \ell = (2M_2 + 1)k - 1, \quad N_0 = \mathsf{W}(M_1, \ell),$$

  showing that $r/(bd) \to \infty$ as $d \to \infty$ for every fixed pair (M_1, M_2), using qualitative finiteness of van der Waerden numbers via the Hales–Jewett theorem [7] without substituting a numerical upper bound for $\mathsf{W}(M_1, \ell)$.

This paper is primarily a sequel to [6, Section 6] in that same algebraic model. We state the model boundary up front: *no multitape Turing machine is known to decompose into linear class-uniform stages*, and none of the results of this paper closes Question (T) or bridges general adaptive or oblivious machines to two-stage linear shared decoders. Within the two-stage linear class-uniform model, however, arithmetic progressions carry an inherent limitation: Behrend's construction [2] produces subsets of $\mathbb{Z}_d$ of size $d/2^{O(\sqrt{\log d})}$ containing no non-trivial 3-term progression, so M_1-colorings with $M_1 = 2^{O(\sqrt{\log d})}$ can avoid even length-3 monochromatic progressions, while even for $M_1 = 2$ monochromatic progressions have length at most $O(\log d)$. Our main results show that replacing arithmetic progressions by *affine Hilbert cubes* (over every field) and by *shifted monochromatic sumsets* (over fields in which d is invertible) removes Ramsey numbers from two-stage

linear shared-decoder composition and yields explicit exponential-cutoff and logarithmic-power rank bounds.

1.2 Summary of results

1. **Ramsey-free cutoff for exact two-stage linear composition (Section 3).** Over an *arbitrary* field F, Lemma 3.1 proves that if a subspace $W \le A$ admits an injective linear readout invariant under the 2^k ordered products $R_{k-1}^{\varepsilon_{k-1}} \cdots R_0^{\varepsilon_0}$ of k endomorphisms $R_0, \dots, R_{k-1} \in \mathrm{End}_F(A)$, then

$$(k+1)\dim_F W \;\le\; \dim_F A \;+\; \sum_{t=0}^{k-1} \dim_F \ker(R_t - \mathrm{id}_A).$$

   Extracting a monochromatic affine M_2-cube with small steps $1 \le s_t \le \lfloor d/(2M_2^2) \rfloor$ by iterated Cauchy–Schwarz (Lemma 3.2) yields Theorem 3.3: whenever two F-linear class-uniform stages K_1, K_2 on $\mathrm{Mat}_{d\times d}(F^b)$ with M_1 and M_2 classes satisfy $K_2 \circ K_1 = \tau$, one has

$$d \;\le\; \max\Big(8M_1M_2^2,\; (2M_1)^{2^{M_2}}\Big).$$

   For fixed M_2, this rules out exact factorizations for growing $M_1(d) = o(d^{1/2^{M_2}})$, including 3-AP-free Behrend partitions with $M_1(d) = 2^{O(\sqrt{\log d})}$. At small constant class counts, however, Theorem 3.3 is weaker than van der Waerden extraction: at $M_1 = M_2 = 2$, Theorem 3.3 gives $d \le 256$, whereas [6, Corollary 6.4] gives $d \le 16$. Two finite controls accompany Theorem 3.3: at $d = 13, q = 2$ on the 3-AP-free 2-cube $J_{13} = \{0,1,4,5\} \subseteq \mathbb{Z}_{13}$, Example 3.5 shows $9 \le \max \dim_F W \le 10$ over fields with $(13 : F) \neq 0$ (refuting a naive 2^k-multiplier bound $\dim_F W \le 8$); and at $d = 4 = M_1M_2$ on the identical two-interval partition $p_1 = p_2 = [0,0,1,1]$, Example 3.6 constructs exact linear coded factorizations $K_2 \circ K_1 = \tau$ for all lane widths b over every field with $(5 : F) \neq 0$ (leaving characteristic 5 open), while showing that entry-copy and bit-lane-copy factorizations fail.

2. **Log-power error-rank lower bound for approximate two-stage composition (Section 4).** Over any field F with $(d : F) \neq 0$, we pass to a splitting field carrying an injective additive character $\chi : \mathbb{Z}_d \to E^\times$ and replace progressions of length $(2M_2+1)k-1$ by $(2M_2)$-fold monochromatic sumsets $\widetilde{J}_0 + \widetilde{B}_1 + \cdots + \widetilde{B}_{2M_2-1}$ with $|\widetilde{J}_0| = k$, $0 \in \widetilde{B}_t$, and $|\widetilde{B}_t| = k+1$. Because $\big(\chi(\nu c)\big)_{c\in\widetilde{J}_0, 0\le\nu<k}$ is a Vandermonde matrix in the distinct scalars $\chi(c)$ $(c \in \widetilde{J}_0)$, stacked evaluation over any k-element set $\widetilde{J}_0$ is injective on the frequency-$< k$ band (Lemma 4.2), while each $(k+1)$-element summand $\widetilde{B}_t$ supports a nonzero polynomial P_t of degree at most $\mathrm{diam}_{\mathbb{N}}(\widetilde{B}_t)$ annihilating the band (Lemma 4.3). Combining the resulting rank inequality $2\,\mathrm{rank}_F(\tau - K_2 \circ K_1) \ge bk\big(d - \sum_t \mathrm{diam}_{\mathbb{N}}(\widetilde{B}_t)\big)$ (Theorem 4.4) with iterated Kővári–Sós–Turán sumset extraction (Lemma 4.5) yields Theorem 4.6: for all $M_1, M_2 \ge 1$ and all $d \ge D_0(M_1, M_2)$ with $(d : F) \neq 0$,

$$\mathrm{rank}_F(\tau - K_2 \circ K_1) \;\ge\; c(M_1, M_2)\, b\, d\, (\log d)^{\frac{1}{2M_2-1}},$$

   and in particular $\Omega(b\,d\log d)$ when $M_2 = 1$ (Corollary 4.8). The bound is asymptotic: the formalized threshold $D_0(M_1, M_2)$ printed in Theorem 4.6 satisfies $D_0(1,1) = 16^{36} + 2 \approx 2.23 \times 10^{43}$. Example 4.9 constructs an explicit 24-dimensional 4-channel band witness over $\mathbb{C}$ at $d = 12$ showing that stage-1 shift invariance cannot be relaxed to an abstract linear readout on a single $k = 2$ band space.

3. **Three-term swap codes and multi-scale interval masks (Section 5).** In [3, Remark 5.6], examples of three-term swap codes $\{X \in \mathrm{Mat}_{d\times d}(F) : \sum_{t=0}^{2} A_t X B_t^{\mathsf{T}} = X^{\mathsf{T}}\}$ exceeding $d(d+1)/2$ were given at $d = 3$ ($8 > 6$, over every field) and $d = 4$ ($11 > 10$, checked numerically over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$), with no example known for $d \geq 5$. In Theorems 5.1 and 5.2, we construct a 3×3 block family on $d = 3m$ achieving exact dimension $4m(m+1)$ over every field—exceeding $d(d+1)/2$ at $d \in \{3, 6, 9, 12\}$—and pad it to obtain $29 > 28$ at $d = 7$ and $11 > 10$ at $d = 4$ over every field; $d = 5$ and every $d \geq 8$ outside $\{9, 12\}$ remain open. On [3, Problem 9.3], Proposition 5.3 records the upper bound $(d^2 + d\min(\beta_A, \beta_B))/2$ obtained by applying [3, Theorem 6.4] to the common interval-endpoint refinement on one side (which uses only one side's sparsity and is vacuous once $\min(\beta_A, \beta_B) \geq d$), while Proposition 5.4 contrasts interval boundaries (≤ 2) with periodic Walsh digit masks (2^{j+1} boundaries, profile 2 per scale, and operator span dimension $\leq (m+1)2^m$).

4. **Restricted shuffle-and-local operator pipelines and anti-diagonal untwisting (Section 6).** Across the row/column tensor cut $F_{\mathrm{row}}^d \otimes_F (F_{\mathrm{col}}^d \otimes_F F^b)$, Operator Schmidt Rank satisfies $\mathrm{rk}_\otimes(\tau) = d^2$ and $\mathrm{rk}_\otimes(S) = \mathrm{rk}_\otimes(S^{-1}) = 4$ for Stone's perfect shuffle and unshuffle [10] on $d = 2^m \geq 2$ (Theorem 6.2), forcing $t \geq m = \log_2 d$ shuffle/unshuffle passes in any pipeline composed solely of S, S^{-1}, row-uniform operators $I_d \otimes B$, and column-uniform operators $A \otimes I_{db}$ (Theorem 6.3). As we emphasize at the start of Section 6, this applies only to that restricted operator pipeline and *does not* answer [3, Problem 9.5]: within-row cyclic shear $X_{r,c} \mapsto X_{r,(c+r) \bmod d}$ has $\mathrm{rk}_\otimes = d$ (Proposition 6.5) yet runs in $O(L)$ time on a 2-work-tape Turing machine (Proposition 6.4, proved in the shared library `lean-machine` alongside 3-tape shuffle and 2-tape unshuffle). Section 6.2 also records the anti-diagonal coordinate isomorphism intertwining rot_t with $\mathrm{id} \otimes \sigma^t$ (Theorem 6.7) and the two-subspace Grassmann relaxation (Proposition 6.8).

5. **Cut-cone evaluation, random-order set capacity, and why column classes cannot absorb per-entry cut labels (Section 7).** On frozen trace/cut fibres of `Transpose.TM`, Lemma 7.1 constructs depth-D symbolic `CausalConeTree`s directly from $M.\delta$ whose syntactic input-cell support has size at most $b(M.k+1)^D$. Theorem 7.3 formalizes in Lean the $1/(\Delta+1)$ random-order acyclic-subgraph bound of [4, Proposition 4.14(a)–(b)] (after [1, Theorem 3]) in the column-class setting. Finally, Remark 7.5 and Proposition 7.6 explain why neither discarding cut-touching outputs nor refining column classes by per-column cut slots yields a non-vacuous sweep bound: causal spine edges $u \to u+1$ force at least $d^2 - D$ outputs to touch the cut, while column-level cut slots multiply the class count exponentially, showing why bridging periodic stretches requires the per-position fixed-alphabet form of [3, Problem 9.4(b)].

1.3 Formalization conventions

Every numbered theorem, lemma, proposition, corollary, and example in this paper is completely proved and kernel-verified in Lean 4 (version `v4.34.1` with Mathlib [11]) across the `BoundedArityDecoders` package and the shared `LeanMachine.ShuffleShear` modules in `lean-machine`, depending solely on Lean's three standard axioms (`propext`, `Classical.choice`, and `Quot.sound`), with zero `sorry`, `admit`, `native_decide`, or custom axioms. Each numbered result carries a `Lean`: status line citing proved `theorem` declarations, followed by an outline of the formal proof.

2 Model and preliminaries

2.1 Matrix states, class-uniform operators, and bounded-arity set codes

For $d \geq 1$, write $\mathbb{Z}_d = \mathbb{Z}/d\mathbb{Z}$ (formalized as `ZModd`). Given an entry alphabet or module A, a $d \times d$ matrix state is a map $X : \mathbb{Z}_d \to \mathbb{Z}_d \to A$, written $X \in \mathrm{Mat}_{d\times d}(A)$ or $A^{d\times d}$ (`SharedDecoders.State`). Its transpose $\tau(X) = X^{\mathsf{T}}$ (`SharedDecoders.swap`, or `SharedDecoders.swapLinear` when A is an F-module) and column-cyclic shift $\mathrm{rot}_c(X)$ for $c \in \mathbb{Z}_d$ (`SharedDecoders.rot`) are defined by

$$\tau(X)_{i,j} \;=\; X_{j,i}, \qquad \mathrm{rot}_c(X)_{i,j} \;=\; X_{i,\,j+c} \qquad (i, j \in \mathbb{Z}_d).$$

Thus column 0 of $\mathrm{rot}_c(X)$ is column c of X, whereas column c of $\tau(X)$ is the vector $r \mapsto X_{c,r}$ (row c of X).

When $A = F^b$ (`Finb->F`) over a field F, a linear operator $K : \mathrm{Mat}_{d\times d}(F^b) \to \mathrm{Mat}_{d\times d}(F^b)$ is *class-uniform* for a column partition $p : \mathbb{Z}_d \to \kappa$ (`SharedDecoders.ClassUniform`; cf. [3, Definition 6.3] and [6, Section 2]) if there exist F-linear class decoders $f_m : \mathrm{Mat}_{d\times d}(F^b) \to (F^b)^{\mathbb{Z}_d}$ ($m \in \kappa$) such that column c of $K(X)$ equals $f_{p(c)}(\mathrm{rot}_c(X))$ for every X and every $c \in \mathbb{Z}_d$. More generally, `SharedDecoders.IsSharedCode` ([6, Definition 2.1]) restricts the code $C \leq \mathrm{Mat}_{d\times d}(A)$ to an F-linear submodule while allowing arbitrary deterministic maps f_m.

For arbitrary subsets $C \subseteq \Sigma^{d\times d}$ over a finite alphabet Σ (used in Section 7), we also record syntactic cell footprints:

Definition 2.1 (Syntactic read footprints and bounded-arity shared set codes)**.** Let $d \geq 1$, let $M, \Delta \in \mathbb{N}$, and let Σ be a finite alphabet.

(a) A family of class decoders $f : \mathrm{Fin}\, M \to \Sigma^{d\times d} \to \Sigma^{\mathbb{Z}_d}$ has *syntactic cell support* $\leq \Delta$ with footprint table $Q : \mathrm{Fin}\, M \to \mathbb{Z}_d \to \mathrm{Finset}(\mathbb{Z}_d \times \mathbb{Z}_d)$ (`SyntacticCellSupport`) if $|Q_{m,r}| \leq \Delta$ for all (m, r), and whenever $Y_1, Y_2 \in \Sigma^{d\times d}$ agree on every cell $q \in Q_{m,r}$, one has $f_m(Y_1)_r = f_m(Y_2)_r$. (Its whole-column analogue is `SyntacticColumnSupport`.)

(b) Given $p : \mathbb{Z}_d \to \mathrm{Fin}\, M$, a broken-output set $\mathcal{E} \subseteq \mathbb{Z}_d \times \mathbb{Z}_d$, and $C \subseteq \Sigma^{d\times d}$, the data $(\Delta, p, \mathcal{E}, f, Q)$ form a *bounded-arity shared set code on* C (`IsBoundedAritySetCode`) if (f, Q) has syntactic cell support $\leq \Delta$ and

$$f_{p(c)}\big(\mathrm{rot}_c(X)\big)_r \;=\; X_{c,r} \qquad \text{for all } X \in C \text{ and all } (r, c) \notin \mathcal{E}.$$

2.2 Multitape Turing machines, canonical words, and grid bridges

In Sections 6 and 7 we interface with the multitape Turing machine model `Transpose.TM` imported unchanged from [5, Definition 2.1]: a machine M has state set $\mathrm{Fin}(M.q + 1)$, work alphabet $\mathrm{Fin}(M.g + 1)$, $M.k$ two-way read-write work tapes, one two-way read-only binary input tape of length $L = d^2 b$ where $b =$ `Transpose.entryBits`(d), and one one-way write-only binary output tape. The canonical word alphabet (`CanonicalWordAlphabet`) is $\Sigma_d := \{0,1\}^b$ (`Fin(Transpose.entryBitsd)->Bool`).

Lemma 2.2 (Canonical word width and dummy-tape padding)**.** Lean: `entryBits_eq_clog` [NativeSweepInterfaceProofs.lean]
Lean: `canonicalWordAlphabet_card_and_ge_two` [SetCapacityConsequences.lean]
Lean: `padDummyWorkTape_transposesWithin` [MachineSweepConsequences.lean]

For every $d \geq 0$, $2 \leq |\Sigma_d|$, and for every $d \geq 2$, `Transpose.entryBits`$(d) = \lceil \log_2 d \rceil$ *and* $|\Sigma_d| = 2^{\lceil \log_2 d \rceil}$*. Moreover, for every time bound $T : \mathbb{N} \to \mathbb{N}$ and every machine M satisfying* `Transpose.TransposesWithin` $M\ T$*, there exists a machine M^+ with $1 \leq M^+.k$ that also satisfies* `Transpose.TransposesWithin` $M^+\ T$*.*

Outline of the formal proof. Since $1 \leq$ `Transpose.entryBits`(d) for all d, $|\Sigma_d| = 2^b \geq 2$, and for $d \geq 2$ the definition $\lfloor \log_2(d-1) \rfloor + 1$ agrees with `Nat.clog` $2\ d$. When $M.k = 0$, adjoining a single blank work tape whose head never moves preserves states, input-head positions, and emitted output bits step for step. □

To connect Boolean tape matrices $A \in$ `Transpose.Matrix` with the linear operators of Section 6, `encodeTMMatrixToGrid` maps each bit $A(r, c, \ell) \in \{0, 1\}$ to $0, 1 \in F$ in `GridState` $F\ d\ b = (\mathrm{Fin}\, d \to \mathrm{Fin}\, d \to \mathrm{Fin}\, b \to F)$.

Proposition 2.3 (Equivariance of the matrix-to-grid bridge)**.** Lean: `tmMatrix_gridLinear_bridges` [ShuffleShearBridge.lean]

For every ring F, `encodeTMMatrixToGrid` *intertwines* `stoneShuffleTMMatrix` *with* `stoneShuffleGridLinear` *(for $d = 2^m$),* `stoneUnshuffleTMMatrix` *with* `stoneUnshuffleGridLinear` *(for $d = 2^m$),* `rowShearTMMatrix` *with* `rowShearLinear` *(for $d \geq 1$), and* `Transpose.transpose` *with* `gridSwapLinear`*.*

Outline of the formal proof. Verified directly by unfolding `encodeTMMatrixToGrid` on each coordinate permutation. □

3 Affine Hilbert cubes and Ramsey-free two-stage cutoffs

3.1 Common-message growth on ordered operator cubes

Suppose two F-linear class-uniform operators $K_1, K_2 : \mathrm{Mat}_{d\times d}(F^b) \to \mathrm{Mat}_{d\times d}(F^b)$ with class partitions $p_1 : \mathbb{Z}_d \to \mathrm{Fin}\, M_1$ and $p_2 : \mathbb{Z}_d \to \mathrm{Fin}\, M_2$ satisfy $K_2 \circ K_1 = \tau$. By the common-message reduction of [6, Lemma 6.1 and Theorem 6.3], any such factorization produces a subspace $W \leq A$ of dimension $\dim_F W = bd$ inside a q-channel cyclic shift module $A = (\mathbb{Z}_d \to F^q)$ with $q \leq bM_2$ ($\dim_F A = qd \leq bM_2 d$), together with a linear readout $L : A \to Z$ injective on W such that $L(T_c w) = L(w)$ for all $w \in W$ and all c in a single p_1-class $J = p_1^{-1}(m_1)$ (where $(T_s w)_u = w_{u+s}$, formalized as `cyclicChannelShift`). In [6, Theorem 6.3], choosing a monochromatic arithmetic progression $\{c_0 + js : 0 \leq j < \ell\} \subseteq J$ gave $\ell \dim_F W \leq \dim_F A + (\ell - 1)\dim_F \ker(T_s - \mathrm{id})$, yielding $d \leq M_2(\mathsf{W}(M_1, M_2 + 1) - 1)$.

To dispense with arithmetic progressions over an arbitrary field F, consider any k linear endomorphisms $R_0, \ldots, R_{k-1} \in \mathrm{End}_F(A)$ (not required to commute) and define for each $\varepsilon \in \{0, 1\}^k$ the ordered operator cube product (`orderedCubeMap`)

$$R^{\varepsilon} \;:=\; R_{k-1}^{\varepsilon_{k-1}} \cdots R_0^{\varepsilon_0} \in \mathrm{End}_F(A).$$

Lemma 3.1 (Affine k-cube common-message growth)**.** Lean: `lemma_AffineCubeCommonMessageGrowth` [`AffineCubeCompositionProofs.lean`]

Let F be any field, A a finite-dimensional F-vector space, Z an F-vector space, $R_0, \dots, R_{k-1} \in \mathrm{End}_F(A)$, $W \le A$ a subspace, and $L : A \to Z$ an F-linear map injective on W. If $L(R^\varepsilon x) = L(x)$ for all $x \in W$ and all $\varepsilon \in \{0,1\}^k$, then

$$(k+1)\dim_F W \;\le\; \dim_F A \;+\; \sum_{t=0}^{k-1} \dim_F \ker(R_t - \mathrm{id}_A).$$

Outline of the formal proof. Define the nested cube-orbit subspaces $S_0 := W$ and $S_{t+1} := S_t + R_t(S_t)$ for $0 \le t < k$. By induction on t, every $u \in S_t$ is a linear combination of elements $R^\varepsilon x$ with $x \in W$ and $\varepsilon_s = 0$ for $s \ge t$. Comparing $\varepsilon_t = 1$ with $\varepsilon_t = 0$ in the cube invariance $L(R^\varepsilon x) = L(x)$ shows that $P_t := R_t - \mathrm{id}_A$ maps S_t into $S_{t+1} \cap \ker L$. Since $W \subseteq S_t \subseteq S_{t+1}$ and L is injective on W, we have $W \cap P_t(S_t) \subseteq W \cap \ker L = \{0\}$, so

$$\dim_F S_{t+1} \;\ge\; \dim_F W + \dim_F P_t(S_t) \;\ge\; \dim_F W + \dim_F S_t - \dim_F \ker(R_t - \mathrm{id}_A).$$

Summing from $t = 0$ to $k-1$ gives $(k+1)\dim_F W \le \dim_F S_k + \sum_{t=0}^{k-1} \dim_F \ker(R_t - \mathrm{id}_A) \le \dim_F A + \sum_{t=0}^{k-1} \dim_F \ker(R_t - \mathrm{id}_A)$. □

3.2 Iterated Cauchy–Schwarz cube extraction and Theorem 2A

Unlike arithmetic progressions, affine k-cubes exist in every dense subset $J \subseteq \mathbb{Z}_d$ (including 3-AP-free sets) by elementary bucket pigeonholing and Cauchy–Schwarz, with explicit step bounds $1 \le s_t \le \lfloor d/(2k^2) \rfloor$ keeping $\dim_F \ker(T_{s_t} - \mathrm{id})$ small.

Lemma 3.2 (Cauchy–Schwarz affine k-cube extraction and dense-class bound)**.** Lean: `cauchySchwarzAffineCubeExtraction` [`AffineCubeCompositionProofs.lean`]

Lean: `singleSubset_CauchySchwarzAffineCubeExtraction`, `denseSubset_AffineCubeCommonMessageBound` [`AffineCubeConsequences.lean`]

Let $d, M_{\mathrm{eff}}, k \ge 1$ satisfy $\max(8M_{\mathrm{eff}}k^2,\ (2M_{\mathrm{eff}})^{2^k}) < d$.

(a) Single-subset form*: for every subset $J \subseteq \mathbb{Z}_d$ of size $|J| \ge d/M_{\mathrm{eff}}$, there exist $c_0 \in J$ and steps $s_0, \dots, s_{k-1} \in \{1, \dots, \lfloor d/(2k^2) \rfloor\}$ such that $c_0 + \sum_{t:\varepsilon_t=1} s_t \in J$ for all $\varepsilon \in \{0,1\}^k$. Over any field F, any subspace $W \le (\mathbb{Z}_d \to F^q)$ admitting a linear readout L and a reference shift $c_{\mathrm{ref}} \in J$ such that $w \mapsto L(T_{c_{\mathrm{ref}}} w)$ is injective on W and $L(T_c w) = L(T_{c_{\mathrm{ref}}} w)$ for all $c \in J$ satisfies $(k+1)\dim_F W \le qd + qk\lfloor d/(2k^2) \rfloor$, ruling out exact two-stage linear factorizations whenever one stage-1 class has size $\ge d/M_{\mathrm{eff}}$ at $k = M_2$ and $b \ge 1$.*

(b) Coloring form*: setting $M_{\mathrm{eff}} = M_1$ and $k = M_2$, every M_1-coloring $p_1 : \mathbb{Z}_d \to \mathrm{Fin}\, M_1$ admits a monochromatic affine M_2-cube with steps in $\{1, \dots, \lfloor d/(2M_2^2) \rfloor\}$.*

Outline of the formal proof. Lift J to $S \subseteq \{0, \dots, d-1\}$. Partitioning $\{0, \dots, d-1\}$ into $Q = 2k^2$ intervals of length $\le \lfloor d/Q \rfloor + 1$ and applying Cauchy–Schwarz to the bucket sizes shows that some step $s \in \{1, \dots, \lfloor d/Q \rfloor\}$ satisfies $|S|(|S| - Q) \le 2d\,|\{x \in S : x + s \in S\}|$. Iterating k times evolves the density denominator by $D_{t+1} = 2D_t^2$ ($D_0 = M_{\mathrm{eff}}$, so $D_k = \frac{1}{2}(2M_{\mathrm{eff}})^{2^k}$); the condition $d > \max(8M_{\mathrm{eff}}k^2, (2M_{\mathrm{eff}})^{2^k})$ keeps every iterate nonempty and prevents wraparound modulo d since $\sum_{t=0}^{k-1} s_t \le k\lfloor d/(2k^2) \rfloor < d$. Feeding the extracted cube into Lemma 3.1 with $\dim_F \ker(T_{s_t} - \mathrm{id}) \le q s_t \le q\lfloor d/(2k^2) \rfloor$ gives the dimension bound. □

Theorem 3.3 (Theorem 2A: Ramsey-free two-stage exact factorization cutoff)**.** Lean: `theorem2A_TwoStageRamseyFreeCutoff` [`AffineCubeCompositionProofs.lean`]
Lean: `theorem2A_GrowingClassesNonFactorization` [`AffineCubeConsequences.lean`]

Let F be any field, let $d, b, M_1, M_2 \geq 1$, and let $p_1 : \mathbb{Z}_d \to \operatorname{Fin} M_1$ and $p_2 : \mathbb{Z}_d \to \operatorname{Fin} M_2$. If two F-linear class-uniform maps K_1, K_2 on $\operatorname{Mat}_{d\times d}(F^b)$ satisfy $K_2 \circ K_1 = \tau$, then

$$d \;\leq\; \max\Big(8M_1M_2^2,\; (2M_1)^{2^{M_2}}\Big).$$

Consequently, for any class schedules $M_1(d), M_2(d)$ eventually satisfying both $8M_1(d)M_2(d)^2 < d$ and $(2M_1(d))^{2^{M_2(d)}} < d$ (for instance, fixed M_2 with $M_1(d) = o\big(d^{1/2^{M_2}}\big)$, covering 3-AP-free Behrend partitions [2] with $M_1(d) = 2^{O(\sqrt{\log d})}$, or fixed M_1 with $M_2(d) \leq (1-\varepsilon)\log_2\log_2 d$), no exact two-stage linear factorization $K_2 \circ K_1 = \tau$ exists over any field F.

Outline of the formal proof. Assume for contradiction that $\max(8M_1M_2^2, (2M_1)^{2^{M_2}}) < d$ and $K_2 \circ K_1 = \tau$. By Lemma 3.2(b), some p_1-class contains an affine M_2-cube with steps $1 \leq s_t \leq \lfloor d/(2M_2^2)\rfloor$. On the q-channel common-message module $A = (\mathbb{Z}_d \to F^q)$ ($q \leq bM_2$, $\dim_F A = qd \leq bM_2d$), [6, Lemma 6.1 and Theorem 6.3] supplies a subspace $W \leq A$ of dimension $\dim_F W = bd$ while $\dim_F \ker(T_{s_t} - \mathrm{id}) \leq qs_t \leq bM_2\lfloor d/(2M_2^2)\rfloor$. Applying Lemma 3.1 with $k = M_2$ gives

$$(M_2+1)bd \;\leq\; bM_2d + \sum_{t=0}^{M_2-1} bM_2\lfloor d/(2M_2^2)\rfloor \;\leq\; bM_2d + \frac{1}{2}bd,$$

contradicting $bd > 0$. □

Remark 3.4 (Comparison with van der Waerden cutoffs at small parameters)**.** Theorem 3.3 trades small-parameter tightness for elementary dependence on M_1. At $M_1 = M_2 = 2$, Theorem 3.3 gives $d \leq \max(64, 4^4) = 256$, which is substantially weaker than the bound $d \leq 2(\mathsf{W}(2,3) - 1) = 16$ of [6, Corollary 6.4]. Conversely, once $M_1 \geq 3$ and $M_2 \geq 3$ grow or p_1 is chosen without 3-term progressions, the van der Waerden bound $M_2(\mathsf{W}(M_1, M_2+1) - 1)$ grows as Hales–Jewett numbers in M_1, M_2, whereas Theorem 3.3 remains polynomial in M_1 for every fixed M_2.

3.3 Finite controls at $d = 13$ and $d = 4$

Two finite-dimensional examples test the linear-growth multiplier $(k+1)$ in Lemma 3.1 and the coded-versus-copy distinction at $d = 4 = M_1M_2$.

Example 3.5 (3-AP-free 2-cube witness at $d = 13$)**.** Lean: `control_D13TwoCubeAPFreeSharpness` [`AffineCubeCompositionProofs.lean`]
Lean: `control_D13_GeneralFieldAndProgressionBridge` [`AffineCubeConsequences.lean`]
Let $d = 13$, $q = 2$ ($\dim_F A = 26$), and consider the 2-cube $J_{13} = \{0,1,4,5\} = \{0,1\} + \{0,4\} \subseteq \mathbb{Z}_{13}$.

(a) J_{13} is 3-AP-free modulo 13 (`ThreeAPFreeMod`), so every non-constant progression in J_{13} has length $\ell \leq 2$, where the single-step progression bound gives only $\dim_F W \leq 14$ (attained on $\{0,1\}$ over every field).

(b) Over every field F, Lemma 3.1 with $k = 2$ and $\dim_F \ker(T_s - \mathrm{id}) = 2$ ($s \in \{1,4\}$) gives $3\dim_F W \leq 26 + 2 + 2 = 30$, hence $\dim_F W \leq 10$.

(c) Over $\mathbb{R}$, and over every field F with $(13 : F) \neq 0$, there exists an explicit (W, L) shift-invariant on J_{13} of dimension $\dim_F W = 9$. Thus the maximum invariant dimension on J_{13} over such fields lies in $\{9, 10\}$; since $4 \cdot 9 = 36 > 26 + 3 \cdot 2 = 32$, a naive 2^k-multiplier inequality $2^k \dim_F W \leq \dim_F A + (2^k - 1)q$ ($\dim_F W \leq 8$) does not hold.

Outline of the formal proof. Residue inspection verifies that J_{13} has no non-constant 3-term progression modulo 13, and cyclic shift periodicity gives $\dim_F \ker(T_s - \mathrm{id}) = 2$ for $s \in \{1, 4\}$. For the 9-dimensional witness, the constant mode contributes 2 dimensions and 7 non-zero Fourier modes each satisfy a rank-3 alignment system inside a 4-dimensional mode space, contributing $2 + 7 = 9$ independent dimensions. □

Example 3.6 (Same-interval coded factorizations at $d = 4 = M_1 M_2$ when $(5 : F) \neq 0$)**.** `Lean: control_D4SameIntervalCodedFactorization` [`AffineCubeCompositionProofs.lean`]
`Lean: control_D4_AllWidthsAndBinaryFields` [`AffineCubeConsequences.lean`]
Let $d = 4$ and let both stages use the same two-interval partition $p_1 = p_2 = [0, 0, 1, 1]$ (`halfIntervalPartition4`, $M_1 = M_2 = 2$).

(a) *Uncoded impossibility*: for every lane width $b \geq 1$ and finite alphabet $|\Sigma| \geq 2$, neither entry-copy rules nor bit-lane-copy rules K_1, K_2 with $p_1 = p_2 = [0, 0, 1, 1]$ can satisfy $K_2 \circ K_1 = \tau$.

(b) *Coded factorization for* $(5 : F) \neq 0$: over every field F with $(5 : F) \neq 0$ (combining an explicit rational table valid whenever $(20 : F) \neq 0$ with a separate binary 0/1 table valid whenever $(2 : F) = 0$, including $\mathbb{F}_2$ at $b = 1$), and for every lane width $b \geq 0$, there exist F-linear class-uniform maps K_1, K_2 on $\mathrm{Mat}_{4\times 4}(F^b)$ with $p_1 = p_2 = [0, 0, 1, 1]$ satisfying $K_2 \circ K_1 = \tau$. (Whether such a factorization exists in characteristic 5 is not decided here.)

Outline of the formal proof. In (a), projecting onto lane 0 reduces bit-lane copy to scalar entry copy, where routing 4 entries through 2-element interval classes forces a collision. In (b), separate explicit linear tables verify $K_2 \circ K_1 = \tau$ lane-by-lane when $(20 : F) \neq 0$ and when $(2 : F) = 0$; every field with $(5 : F) \neq 0$ satisfies one of the two conditions. □

4 Shifted sumsets and logarithmic two-stage rank deficits

4.1 Minimum-shifted monochromatic sumsets

For approximate two-stage compositions at fixed class counts $M_1, M_2 \geq 1$, write

$$r \;:=\; \mathrm{rank}_F(\tau - K_2 \circ K_1).$$

In [6, Theorem 6.6], extracting a monochromatic progression of length $\ell = (2M_2 + 1)k - 1$ inside p_1 gave $r \geq bk(d - N_0 + 1)/2$ with $N_0 = \mathsf{W}(M_1, \ell)$ over every field F with $(d : F) \neq 0$, proving $r/(bd) \to \infty$ as $d \to \infty$. Because N_0 is a van der Waerden number in $(2M_2 + 1)k - 1$, however, the resulting factor $k(d)$ grows much more slowly than any iterated logarithm of d.

Over any field F in which d is invertible ($(d : F) \neq 0$), we pass to a splitting field E/F carrying an injective additive character $\chi : \mathbb{Z}_d \to E^\times$ and replace progressions by $(2M_2)$-fold monochromatic sumsets

$$J_0 + B_1 + \cdots + B_{2M_2 - 1} \;\subseteq\; p_1^{-1}(m_1), \qquad |J_0| = k, \quad |B_t| = k + 1.$$

For any finite $S \subseteq \mathbb{N}$, write $\mathrm{diam}_{\mathbb{N}}(S) := \max S - \min S$ (0 if $S = \emptyset$; `finsetNatDiam`). Shifting each B_t down by $\min B_t$ places $0 \in \widetilde{B}_t$ while preserving $\mathrm{diam}_{\mathbb{N}}(\widetilde{B}_t) = \mathrm{diam}_{\mathbb{N}}(B_t)$, so all partial subset sums remain in $p_1^{-1}(m_1)$ simultaneously.

Lemma 4.1 (Minimum-shifting monochromatic sumsets)**.** Lean: `unshiftedToShiftedSumset` [`SumsetRankDeficitProofs.lean`]
Lean: `finsetNatDiam_shift_min`, `unshiftedToShiftedSumset_intrinsicDiam` [`SumsetRankConsequences.lean`]

Let $d, M_1 \geq 1$ and $p_1 : \mathbb{Z}_d \to \operatorname{Fin} M_1$. Given any unshifted monochromatic sumset $J_0 + B_1 + \cdots + B_n \subseteq p_1^{-1}(m_1)$ with $|J_0| = k$, $|B_t| = k+1$, and $B_t \subseteq [0, L_0]$, setting $\widetilde{B}_t := B_t - \min B_t$ and $\widetilde{J}_0 := J_0 + \sum_{t=1}^{n} \min B_t$ satisfies $0 \in \widetilde{B}_t$, $|\widetilde{B}_t| = k+1$, $\operatorname{diam}_{\mathbb{N}}(\widetilde{B}_t) = \operatorname{diam}_{\mathbb{N}}(B_t) \leq L_0$, $|\widetilde{J}_0| = k$, and

$$\widetilde{J}_0 + \sum_{t \in S} \widetilde{B}_t \;\subseteq\; p_1^{-1}(m_1) \qquad \text{for every subset } S \subseteq \{1, \ldots, n\}$$

(packaged directly at $n = 2M_2 - 1$ for $M_2 \geq 1$ in `unshiftedToShiftedSumset_intrinsicDiam`*).*

Outline of the formal proof. Choosing $b_t \in \widetilde{B}_t$ for $t \in S$ and $0 \in \widetilde{B}_t$ for $t \notin S$ rewrites $c_0 + \sum_t \min B_t + \sum_{t \in S} b_t = c_0 + \sum_{t=1}^{n} (b_t + \min B_t)$ with $b_t + \min B_t \in B_t$. Translation by $-\min B_t$ on $\mathbb{N}_{\geq \min B_t}$ preserves both cardinality and $\max - \min$. □

4.2 Arbitrary-subset Vandermonde pullback and sparse annihilating polynomials

On an arbitrary k-element subset $J_0 \subseteq \mathbb{Z}_d$, the character-evaluation matrix $\big(\chi(\nu \cdot c)\big)_{c \in J_0,\, 0 \leq \nu < k}$ is a $k \times k$ Vandermonde matrix in the k distinct scalars $z_c = \chi(c)$ ($c \in J_0$), since $\chi(\nu \cdot c) = \chi(c)^{\nu}$. Thus J_0 need not be an interval or progression for stacked evaluation on the frequency-$< k$ band to be injective.

Lemma 4.2 (Arbitrary-subset Vandermonde injectivity and stacked band pullback)**.** Lean: `vandermondeArbitrarySubsetInjective`, `stackedBandPullbackInjective` [`SumsetRankDeficitProofs.lean`]

Let $d \geq 1$, let E be a field with an injective additive character $\chi : \mathbb{Z}_d \to E^{\times}$, and let $J_0 \subseteq \mathbb{Z}_d$ satisfy $|J_0| = k \leq d$.

(a) For any E-vector space V, degree-$< k$ character evaluation $a \mapsto \big(\sum_{\nu=0}^{k-1} \chi(\nu c)\, a_\nu\big)_{c \in J_0}$ (`charBandEval`*) is injective on V^k.*

(b) Let $M_1, M_2 \geq 1$, $p_1 : \mathbb{Z}_d \to \operatorname{Fin} M_1$, and $p_2 : \mathbb{Z}_d \to \operatorname{Fin} M_2$. For any two E-linear class-uniform stages K_1, K_2 on $\operatorname{Mat}_{d \times d}(E^b)$ with $J_0 \subseteq p_1^{-1}(m_1)$ and stage-1 class-m_1 decoder f_{1,m_1}, the frequency-$< k$ band space H (`SharedDecoders.bandRowSpace` *at (χ, p_2, k), with $\dim_E H \leq bM_2k$) carries a pullback subspace $W \leq$ `SharedDecoders.ColumnAmbient`(H) satisfying*

$$\dim_E W + \operatorname{rank}_E(\tau - K_2 \circ K_1) \;\geq\; bkd$$

such that the stacked evaluation $x \mapsto \big(f_{1,m_1}(\operatorname{rot}_c(\iota_H x))\big)_{c \in J_0}$ is injective on W.

Outline of the formal proof. In (a), $\chi(\nu c) = \chi(c)^{\nu}$, and injectivity of χ makes $\{\chi(c) : c \in J_0\}$ k distinct scalars in E, so a degree-$< k$ vector-valued polynomial vanishing on J_0 is zero. Part (b) restricts $\tau - K_2 \circ K_1$ to inputs whose columns lie in the k-frequency character subspace (bkd dimensions), intersects with $\ker(\tau - K_2 \circ K_1)$ losing at most $r = \operatorname{rank}_E(\tau - K_2 \circ K_1)$ dimensions, and applies (a) to deduce that vanishing of $K_1 X$ on columns $c \in J_0$ forces $(\tau X)_{:,c} = (K_2 K_1 X)_{:,c} = 0$ on J_0 and hence $X = 0$. □

Each summand $\widetilde{B}_t$ of size $k+1$ provides $k+1$ monomials against k frequencies $\nu \in \{0, \ldots, k-1\}$, guaranteeing a nonzero sparse polynomial supported in $\widetilde{B}_t$ that annihilates the frequency-$< k$ band and has degree at most $\operatorname{diam}_{\mathbb{N}}(\widetilde{B}_t)$.

Lemma 4.3 (Sparse sumset annihilating polynomial)**.** Lean: `sumsetAnnihilatingPolynomial`, `cyclicChannelShift_aeval_ker_finrank_le` [`SumsetRankDeficitProofs.lean`]
Lean: `sumsetAnnihilatingPolynomial_bandRowSpace` [`SumsetRankConsequences.lean`]

Let $d \geq 1$, let E be a field with an injective additive character $\chi : \mathbb{Z}_d \to E^\times$, and let $\widetilde{B} \subseteq \mathbb{N}$ satisfy $0 \in \widetilde{B}$, $|\widetilde{B}| = k+1$, and $\max \widetilde{B} \leq L_0$. Then there exists a nonzero polynomial $P \in E[X]$ supported in $\widetilde{B}$ with $\deg P \leq \operatorname{diam}_{\mathbb{N}}(\widetilde{B}) \leq L_0$ such that $\sum_{x \in \widetilde{B}} P_x \chi(\nu x) = 0$ for all $0 \leq \nu < k$. Moreover, for any $q \in \mathbb{N}$, on the q-channel cyclic shift module $(\mathbb{Z}_d \to E^q)$ one has

$$\dim_E \ker P(T_1) \;\leq\; q \deg P \;\leq\; q\, L_0,$$

and on `SharedDecoders.ColumnAmbient` *over* `SharedDecoders.bandRowSpace` *for (χ, p_2, k) (where $\dim_E H \leq bM_2k$) one has $\dim_E \ker P(T_{\mathrm{band}}) \leq bM_2kL_0$.*

Outline of the formal proof. The homogeneous linear system $\sum_{x \in \widetilde{B}} P_x \chi(\nu x) = 0$ $(0 \leq \nu < k)$ has k equations in $k+1$ unknowns, hence a nonzero solution P supported in $\widetilde{B}$ of degree $\leq \max \widetilde{B} = \operatorname{diam}_{\mathbb{N}}(\widetilde{B}) \leq L_0$. On $(\mathbb{Z}_d \to E^q)$, the linear recurrence $P(T_1)w = 0$ determines w from at most $\deg P$ coordinates once $\deg P \leq d$ (and trivially $\dim \leq qd \leq q \deg P$ when $\deg P > d$). □

4.3 Sumset rank bound, Kővári–Sós–Turán extraction, and Theorem 2B

Theorem 4.4 (Two-stage sumset recurrence rank bound)**.** Lean: `sumsetTwoHopRankBound` [`SumsetRankDeficitProofs.lean`]

Let F be a field with $(d : F) \neq 0$, let $M_1, M_2, b \geq 1$, and let $1 \leq k \leq d$. Suppose $p_1 : \mathbb{Z}_d \to \operatorname{Fin} M_1$ admits a shifted $(2M_2)$-fold monochromatic sumset $(\widetilde{J}_0, \widetilde{B}, \mathrm{diamB})$ in class m_1 with $\sum_{t=0}^{2M_2-2} \mathrm{diamB}_t \leq d$. Then for every $p_2 : \mathbb{Z}_d \to \operatorname{Fin} M_2$ and every pair of F-linear class-uniform maps K_1, K_2 on $\mathrm{Mat}_{d\times d}(F^b)$, writing $r := \operatorname{rank}_F(\tau - K_2 \circ K_1)$,

$$2r \;\geq\; bk\Big(d - \sum_{t=0}^{2M_2-2} \mathrm{diamB}_t\Big), \qquad \text{and if } \mathrm{diamB}_t \leq \lfloor d/(4M_2) \rfloor \text{ for all } t, \quad 4r \;\geq\; bkd.$$

Outline of the formal proof. Since $(d : F) \neq 0$, $X^d - 1$ is separable over F; passing to its splitting field E/F preserves $\operatorname{rank}(\tau - K_2 \circ K_1)$ and furnishes an injective character $\chi : \mathbb{Z}_d \to E^\times$. Let $W \leq A_{\mathrm{band}}$ $(\dim_E A_{\mathrm{band}} \leq bM_2kd)$ be the pullback subspace from Lemma 4.2(b) on $\widetilde{J}_0$, with $\dim_E W \geq bkd - r$, and let P_t be the sparse annihilating polynomial on $\widetilde{B}_t$ from Lemma 4.3. Because P_t annihilates the frequency-$< k$ band while every shift in $\widetilde{J}_0 + \sum_{s \leq t} \widetilde{B}_s$ lies in $p_1^{-1}(m_1)$, $P_t(T_1)$ maps the t-th partial sumset span U_t into the kernel of stacked $\widetilde{J}_0$-evaluation, which meets W trivially. Each of the $2M_2 - 1$ steps therefore adds at least $\dim_E W - bM_2k\,\mathrm{diamB}_t$ dimensions inside A_{band}:

$$2M_2 \dim_E W - bM_2k \sum_{t=0}^{2M_2-2} \mathrm{diamB}_t \;\leq\; \dim_E A_{\mathrm{band}} \;\leq\; bM_2kd.$$

Substituting $\dim_E W \geq bkd - r$ and dividing by M_2 yields $2r \geq bk(d - \sum_t \mathrm{diamB}_t)$, and $\mathrm{diamB}_t \leq \lfloor d/(4M_2) \rfloor$ gives $\sum_t \mathrm{diamB}_t \leq d/2$, hence $4r \geq bkd$. □

Lemma 4.5 (Iterated Kővári–Sós–Turán monochromatic sumset extraction)**.** Lean: `iteratedKovariSosTuranSumset` [`SumsetRankDeficitProofs.lean`]

For every $M_1, M_2 \geq 1$, there exist $0 < c_{\mathrm{lo}} \leq C_{\mathrm{hi}}$ and $D_0 \in \mathbb{N}$ such that for all $d \geq D_0$, every M_1-coloring $p_1 : \mathbb{Z}_d \to \operatorname{Fin} M_1$ contains an unshifted (and hence shifted) $(2M_2)$-fold monochromatic sumset in $[0, \lfloor d/(4M_2) \rfloor]$ at width scale $1 \leq k \leq d$ satisfying

$$c_{\mathrm{lo}} \, (\log d)^{\frac{1}{2M_2-1}} \;\leq\; k \;\leq\; C_{\mathrm{hi}} \, (\log d)^{\frac{1}{2M_2-1}}.$$

Outline of the formal proof. Select the densest color class m_1 inside $[0, \lfloor d/2 \rfloor)$ and form a bipartite addition graph between shifts in $[0, \lfloor d/(4M_2) \rfloor]$ and base positions. Iterating the Kővári–Sós–Turán double count [9] $2M_2 - 1$ times extracts k base elements and $2M_2 - 1$ shift sets of size $k+1$ at density cost exponential in k^{2M_2-1}, balancing at $k = \Theta_{M_1,M_2}\big((\log d)^{1/(2M_2-1)}\big)$. □

Theorem 4.6 (Theorem 2B: log-power two-stage rank deficit over admissible fields)**.** Lean: `theorem2B_TwoStageLogPowerRankDeficit` [`SumsetRankDeficitProofs.lean`]

Lean: `theorem2B_ExplicitConstant`, `theorem2BExplicitConst_pos`, `theorem2BExplicitThreshold_eq_pow` [`SumsetRankConsequences.lean`]

For all integers $M_1, M_2 \geq 1$, define the explicit constant and dimension threshold (`theorem2BExplicitConst`*,* `theorem2BExplicitThreshold`*)*

$$\begin{aligned} c(M_1, M_2) \;&:=\; \frac{1}{144 \, \big(\log(16 M_1 M_2)\big)^{\frac{1}{2M_2-1}}} \;>\; 0, \\ D_0(M_1, M_2) \;&:=\; \max\Big(4M_2,\; \big\lceil \exp\big(36^{2M_2-1} \log(16 M_1 M_2)\big) \big\rceil + 2\Big) \\ &=\; \max\Big(4M_2,\; (16 M_1 M_2)^{36^{2M_2-1}} + 2\Big). \end{aligned}$$

Then for every field F, every $d \geq D_0(M_1, M_2)$ with $(d : F) \neq 0$, every lane width $b \geq 1$, and every pair of F-linear class-uniform stages K_1, K_2 on $\mathrm{Mat}_{d\times d}(F^b)$ with M_1 and M_2 classes,

$$\operatorname{rank}_F(\tau - K_2 \circ K_1) \;\geq\; c(M_1, M_2) \, b \, d \, (\log d)^{\frac{1}{2M_2-1}}.$$

Outline of the formal proof. Instantiate the Kővári–Sós–Turán chain of Lemma 4.5 at $k = \big\lfloor \big(\frac{\log d}{18^{2M_2-1} \log(16 M_1 M_2)}\big)^{1/(2M_2-1)} \big\rfloor \geq 2$ for $d \geq D_0(M_1, M_2)$, and substitute into $4r \geq bkd$ from Theorem 4.4. □

Remark 4.7 (Asymptotic threshold and characteristic restriction)**.** Two limitations of Theorem 4.6 should be noted explicitly. First, the formalized threshold $D_0(M_1, M_2)$ is purely asymptotic: already at $M_1 = M_2 = 1$ one has $D_0(1,1) = 16^{36} + 2 = 2^{144} + 2 \approx 2.23 \times 10^{43}$, and for larger M_2 the exponent 36^{2M_2-1} grows exponentially in M_2. Second, like [6, Theorem 6.6] (and unlike Theorem 3.3), Theorem 4.6 requires $(d : F) \neq 0$ (excluding $\mathbb{F}_2$ at $d = 2^m$) so that $\mathbb{Z}_d$ admits d distinct character values.

Corollary 4.8 (Single-class second stage: $\Omega(b\,d\log d)$ rank deficit)**.** Lean: `theorem2B_OneSecondStage_OmegaLog` [`SumsetRankConsequences.lean`]

When $M_2 = 1$, for every $M_1 \geq 1$ there exist $c > 0$ and $D_0 \geq 2$ such that for every field F, every $d \geq D_0$ with $(d : F) \neq 0$, every $b \geq 1$, and every M_1-class K_1 and 1-class K_2 on $\mathrm{Mat}_{d\times d}(F^b)$,

$$\mathrm{rank}_F(\tau - K_2 \circ K_1) \;\geq\; c\,b\,d\log d.$$

Outline of the formal proof. Specialize Theorem 4.6 to $M_2 = 1$, where $\frac{1}{2M_2-1} = 1$. □

4.4 Abstract band relaxation control at $d = 12$

Theorem 4.4 uses stage-1 class-uniformity across all shifts of a $(2M_2)$-fold sumset rather than relaxing stage 1 to an arbitrary linear readout L on the q-channel ambient space `SharedDecoders.ColumnAmbient` of H over a single half-density subset $J \subseteq \mathbb{Z}_d$ and a single $k = 2$ band. The following 4-channel witness at $d = 12$ shows that this abstract relaxation admits full kd-dimensional solutions.

Example 4.9 (Abstract 4-channel band witness at $d = 12$ over $\mathbb{C}$)**.** Lean: `control_D12SingleClassAbstractBand` [`SumsetRankDeficitProofs.lean`]

Lean: `exists_primitive_twelfth_root_complex`, `d12AddChar_injective_and_primitive`, `control_D12_UnconditionalExists`, `control_D12_BandRowSpaceChannelEquiv` [`SumsetRankConsequences.lean`]

At $d = 12$ and $k = 2$ on the 4-channel cyclic module $(\mathbb{Z}_{12} \to \mathbb{C}^4)$ ($q = 4$, corresponding to scalar lane width $b = 1$ with $M_2 = 2, k = 2$ where $24 = bkd$, or via `control_D12_BandRowSpaceChannelEquiv` to `SharedDecoders.ColumnAmbient` over `SharedDecoders.bandRowSpace` for $(\chi, p_0, 2)$ at $b = 2, M_2 = 1, k = 2$ when L reads 12 scalar coordinates per shift), let $J_{12} = \{0, 1, 3, 4, 7, 9\} \subseteq \mathbb{Z}_{12}$ ($|J_{12}| = 6$). With $\zeta_{12} = \exp(2\pi i/12) \in \mathbb{C}$, there exist a linear readout $L : (\mathbb{Z}_{12} \to \mathbb{C}^4) \to \mathbb{C}^{12}$ and a 24-dimensional subspace $W \leq (\mathbb{Z}_{12} \to \mathbb{C}^4)$ on which stacked evaluation over J_{12} is injective and realizes every pair $(a_0, a_1) \in (\mathbb{C}^{12})^2$ as $L(T_c w) = a_0 + \zeta_{12}^c a_1$ for all $c \in J_{12}$.

Outline of the formal proof. Verify that $\zeta_{12} = \exp(2\pi i/12)$ has exact multiplicative order 12 in $\mathbb{C}$, construct the explicit 24-dimensional 4-channel encoding whose shifts on J_{12} output $a_0 + \zeta_{12}^c a_1$, and identify $(\mathbb{Z}_{12} \to \mathbb{C}^4)$ shift-equivariantly with `SharedDecoders.ColumnAmbient` over `SharedDecoders.bandRowSpace` for $(\chi, p_0, 2)$ at $b = 2, M_2 = 1, k = 2$. Because $q = 4$ corresponds to $bM_2 = 2$ rather than $b = M_2 = 1$, this example isolates the failure of the abstract band relaxation when stage-1 shift covariance is dropped, rather than providing an actual $b = 1, M_1 = M_2 = 1$ two-stage factorization. □

5 Three-term swap codes and multi-scale interval masks

5.1 Three-term swap codes exceeding $d(d+1)/2$ at $d \in \{6, 7, 9, 12\}$ and over all fields at $d = 4$

For finite matrix families $A, B : \alpha \to \mathrm{Mat}_{d\times d}(F)$ over a field F, the swap code (`SwapRigidity.swapCode`) is the solution space

$$\texttt{SwapRigidity.swapCode}(A, B) \;:=\; \Big\{X \in \mathrm{Mat}_{d\times d}(F) \;:\; \sum_{a\in\alpha} A_a X B_a^{\mathsf{T}} = X^{\mathsf{T}}\Big\}.$$

In [3, Theorem 5.1 and Corollary 5.5] (where the coordinate form of Theorem 5.1 is Lean-verified and Corollary 5.5 is proved in text), any two-term swap code ($|\alpha| = 2$) has dimension at most the

symmetric pairing value $d(d+1)/2$, proved by triangularizing a matrix pencil over an algebraic closure. Immediately afterward, [3, Remark 5.6] noted that three terms can exceed $d(d+1)/2$, giving an explicit three-term family of dimension $8 > 6$ at $d = 3$ over every field and an example of dimension $11 > 10$ at $d = 4$ (checked numerically there over $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$), and remarked: *"We have no three-term example for $d \geq 5$."*

Here we give explicit three-term examples above $d(d+1)/2$ at $d \in \{6, 7, 9, 12\}$ over every field F, together with a uniform proof over every field at $d = 4$, by lifting the 3×3 skew-pair construction into $m \times m$ blocks (`threeTermBlockA`, `threeTermBlockB`) and adjoining a diagonal tail (`threeTermPaddedA`, `threeTermPaddedB`). Write the three off-diagonal pairs of $\{0, 1, 2\}$ as $(i_0, j_0) = (0, 1)$, $(i_1, j_1) = (0, 2)$, and $(i_2, j_2) = (1, 2)$, and set

$$A_t \;:=\; (E_{i_t,j_t} - E_{j_t,i_t}) \otimes I_m, \qquad B_t \;:=\; -A_t \;=\; (E_{j_t,i_t} - E_{i_t,j_t}) \otimes I_m \qquad (t \in \{0,1,2\}).$$

Because $E_{j_t,i_t}^{\mathsf{T}} - E_{i_t,j_t}^{\mathsf{T}} = E_{i_t,j_t} - E_{j_t,i_t}$, we have $B_t^{\mathsf{T}} = A_t$, so the swap-code equation $\sum_{t=0}^{2} A_t X B_t^{\mathsf{T}} = X^{\mathsf{T}}$ is

$$\sum_{t=0}^{2} \big((E_{i_t,j_t} - E_{j_t,i_t}) \otimes I_m\big)\, X\, \big((E_{i_t,j_t} - E_{j_t,i_t}) \otimes I_m\big) \;=\; X^{\mathsf{T}}.$$

Theorem 5.1 (Dimension $4m(m+1)$ of the three-term block family on $d = 3m$)**.** Lean: `bonus1_ThreeTermBlockSwapCodeDim` [`SwapCodeAnswersProofs.lean`]

Let F be any field and let $m \geq 1$. Partition $X \in \mathrm{Mat}_{3m\times 3m}(F)$ into a 3×3 grid of $m \times m$ blocks $(X_{i,j})_{0\leq i,j<3}$.

(a) X lies in `SwapRigidity.swapCode`$(A, B) = \{X : \sum_{t=0}^{2} A_t X B_t^{\mathsf{T}} = X^{\mathsf{T}}\}$ *if and only if every block $X_{i,j}$ is symmetric ($X_{i,j}^{\mathsf{T}} = X_{i,j}$ for all $0 \leq i, j < 3$) and the three diagonal blocks sum to zero ($X_{0,0} + X_{1,1} + X_{2,2} = 0$;* `IsThreeByThreeSymmetricZeroTraceBlocks`*).*

(b) Consequently, over every field F,

$$\dim_F \texttt{SwapRigidity.swapCode}(A, B) \;=\; 8 \cdot \frac{m(m+1)}{2} \;=\; 4m(m+1).$$

Since $4m(m+1) - \frac{3m(3m+1)}{2} = \frac{m(5-m)}{2}$, this exceeds the pairing value $d(d+1)/2$ ($d = 3m$) for $m \in \{1, 2, 3, 4\}$ ($d = 3 : 8 > 6$, matching [3, Remark 5.6]; $d = 6 : 24 > 21$; $d = 9 : 48 > 45$; and $d = 12 : 80 > 78$).

Outline of the formal proof. For each $t \in \{0, 1, 2\}$, multiplying on both sides by $(E_{i_t,j_t} - E_{j_t,i_t}) \otimes I_m$ places $+X_{j_t,i_t}$ in block (i_t, j_t), $+X_{i_t,j_t}$ in block (j_t, i_t), $-X_{j_t,j_t}$ in block (i_t, i_t), and $-X_{i_t,i_t}$ in block (j_t, j_t). Summing over the three pairs sends each off-diagonal block (i, j) $(i \neq j)$ to $+X_{j,i}$ and each diagonal block (i, i) to $-\sum_{j\neq i} X_{j,j}$. Equating with $(X_{j,i}^{\mathsf{T}})_{i,j}$ gives $X_{j,i} = X_{j,i}^{\mathsf{T}}$ for $i \neq j$ together with $-X_{1,1} - X_{2,2} = X_{0,0}^{\mathsf{T}}$, $-X_{0,0} - X_{2,2} = X_{1,1}^{\mathsf{T}}$, and $-X_{0,0} - X_{1,1} = X_{2,2}^{\mathsf{T}}$. Over every ring (including characteristic 2), subtracting pairs of diagonal equations yields $X_{0,0}^{\mathsf{T}} - X_{1,1}^{\mathsf{T}} = X_{1,1} - X_{0,0}$ and cyclic analogues, which together with $X_{2,2} = -X_{0,0}^{\mathsf{T}} - X_{1,1}$ is equivalent to $X_{i,i}^{\mathsf{T}} = X_{i,i}$ for $i \in \{0, 1, 2\}$ and $X_{0,0} + X_{1,1} + X_{2,2} = 0$. Eight independent symmetric $m \times m$ blocks (6 off-diagonal and 2 free diagonal blocks) each have dimension $m(m+1)/2$, giving $4m(m+1)$. □

Embedding the $3m \times 3m$ family in the top-left corner of $\mathrm{Mat}_{d\times d}(F)$ ($d = 3m + s$) and setting both A_2 and B_2 equal to I_s on the trailing $s \times s$ diagonal block extends the construction to $d = 4$ and $d = 7$.

Theorem 5.2 (Padded three-term swap codes across $d \in \{3, 4, 6, 7, 9, 12\}$)**.** Lean: `bonus1_ThreeTermPaddedSevenAndFive` [`SwapCodeAnswersProofs.lean`]
Lean: `bonus1_UnifiedThreeTermExceedsPairingBound` [`DualityAndSwapConsequences.lean`]

Over every field F:

(a) Padding $m = 2$ ($3m = 6$) to $d = 7$ achieves dimension $29 > 28 = \frac{7\cdot 8}{2}$; padding $m = 1$ ($3m = 3$) to $d = 4$ achieves $11 > 10 = \frac{4\cdot 5}{2}$; and padding $m = 1$ to $d = 5$ achieves $15 = \frac{5\cdot 6}{2}$.

(b) Consequently, for every field F and every $d \in \{3, 4, 6, 7, 9, 12\}$, there exist $A, B : \mathrm{Fin}\, 3 \to \mathrm{Mat}_{d\times d}(F)$ with

$$\dim_F \texttt{SwapRigidity.swapCode}(A, B) \;>\; \frac{d(d+1)}{2}.$$

Outline of the formal proof. On the trailing $s = d - 3m$ coordinates, A_2 and B_2^{T} both act as $+I_s$, while A_0, B_0, A_1, B_1 vanish outside the top-left $3m \times 3m$ block. Inspecting the coupled cross-blocks between the active $(1, 2)$ pair and the s-coordinate tail yields $24+5 = 29$ dimensions at $(m, d) = (2, 7)$, $8 + 3 = 11$ at $(1, 4)$, and $8 + 7 = 15$ at $(1, 5)$. Combining $(1, 4)$ and $(2, 7)$ with $m \in \{1, 2, 3, 4\}$ in Theorem 5.1 covers $d \in \{3, 4, 6, 7, 9, 12\}$. □

More generally, at $d = 3m + s$ this block-and-tail construction achieves dimension $4m(m + 1) + \frac{s(s+1)}{2} + 2ms$, which is at most $d(d+1)/2$ for every other $d \geq 5$ (giving $15 = 15$ at $d = 5$, $35 < 36$ at $d = 8$, $55 = 55$ at $d = 10$, $63 < 66$ at $d = 11$, $120 = 120$ at $d = 15$, and strictly less than $d(d + 1)/2$ once $m \geq 6$). Thus whether a three-term swap code can exceed $d(d + 1)/2$ remains open for $d = 5$ and for every $d \geq 8$ outside $\{9, 12\}$.

5.2 An observation on multi-scale interval masks and Walsh digit masks (Problem 9.3)

[3, Problem 9.3] asks for upper bounds on $\dim_F \texttt{SwapRigidity.swapCode}(A, B)$ when $d = 2^m$ and both A_a and B_a lie in the span of dyadic masks across all scales composed with cyclic rotations ([3, Corollary 6.13] covers one-sided tensor-factor rules commuting with a single non-derogatory matrix on one address digit). We record a simple observation separating contiguous interval masks from periodic Walsh digit masks.

For contiguous cyclic or aligned dyadic intervals (`IsCyclicInterval`, `IsAlignedDyadicInterval`; spans `InMultiscaleIntervalRotSpan`, `InAlignedDyadicIntervalRotSpan`), each interval mask introduces at most two boundary transitions around $\mathbb{Z}_d$, so applying Theorem TP ([3, Theorem 6.4]) to the common refinement of the interval endpoints on one side at a time gives a nontrivial upper bound whenever at least one side uses $o(d)$ intervals in total:

Proposition 5.3 (One-sided endpoint-refinement bound for multi-scale interval masks)**.** Lean: `bonus2_MultiscaleIntervalBothSides`, `bonus2_MultiscaleFullProfileAttainsFullSwap` [`SwapCodeAnswersProofs.lean`]
Lean: `bonus2_SublinearProfileAndTendstoHalf` [`DualityAndSwapConsequences.lean`]

Let F be any field, $m \geq 0$, and $d = 2^m$.

(a) Every `InAlignedDyadicIntervalRotSpan` family lies in `InMultiscaleIntervalRotSpan`, and whenever finite families $A, B : \alpha \to \mathrm{Mat}_{d\times d}(F)$ lie in `InMultiscaleIntervalRotSpan` with per-scale interval counts $M_A, M_B : \mathrm{Fin}\, m \to \mathbb{N}$, setting $\beta_A := \max(1,\, 2\sum_j M_A(j))$ and $\beta_B := \max(1,\, 2\sum_j M_B(j))$,

$$2 \dim_F \texttt{SwapRigidity.swapCode}(A, B) \;\leq\; d^2 + d \min(\beta_A, \beta_B).$$

(b) Along any sequence of families with $\min(\sum_j M_{A,m}(j), \sum_j M_{B,m}(j))/2^m \to 0$ *as* $m \to \infty$*, for every* $\varepsilon > 0$ *one eventually has* $\dim_{F_m} \texttt{SwapRigidity.swapCode}(A_m, B_m) \le (\frac{1}{2} + \varepsilon)d^2$*.*

(c) At the full aligned dyadic profile $M_{\mathrm{full}}(j) = 2^{j+1}$ *(where* $\sum_{j=0}^{m-1} 2^{j+1} = 2d - 2 \ge d$ *for* $m \ge 1$*), aligned dyadic interval-rotation families realize all matrix units and attain* $\dim_F \texttt{SwapRigidity.swapCode}(A, B) = d^2$*.*

Outline of the formal proof. In (a), the intervals used in A create at most $2\sum_j M_A(j)$ endpoint cuts around $\mathbb{Z}_d$, partitioning $\mathrm{Fin}\, d$ into at most β_A contiguous classes on each of which every A_a acts as a fixed linear combination of rotations. Hence $\sum_a A_a X B_a^{\mathsf{T}} = X^{\mathsf{T}}$ defines a β_A-class-uniform code, giving $2\dim_F \le d^2 + d\beta_A$ by [3, Theorem 6.4]. Passing to $Y = X^{\mathsf{T}}$ exchanges A and B via $Y^{\mathsf{T}} = \sum_a B_a Y A_a^{\mathsf{T}}$, giving $2\dim_F \le d^2 + d\beta_B$ as well. Part (b) divides by $2d^2$. In (c), aligned dyadic intervals down to length $2^0 = 1$ span all diagonal projectors $E_{r,r}$ and hence all matrix units $E_{r,c} = E_{r,r}\sigma^{c-r}$. □

Note the scope of Proposition 5.3: the bound uses only the sparser of the two sides rather than any two-sided interaction between A and B, carries no matching sparse lower bound, and becomes vacuous ($\ge d^2$) as soon as $\min(\beta_A, \beta_B) \ge d$—in particular at the full profile $M_{\mathrm{full}}(j) = 2^{j+1}$ for $m \ge 2$, where $\beta = 4d - 4 > d$.

By contrast, if "scale-j dyadic mask" means the periodic Walsh/Rademacher bit projector $P_{W_j,b}$ onto $\{r \in \mathrm{Fin}(2^m) : \mathrm{bit}_j(r) = b\}$ (`walshBitSet`, `walshDigitMaskRotMatrix`, `IsPeriodicScaleDyadicMask`), then a single mask per scale has exponentially many boundaries at fine scales, so one-sided endpoint refinement becomes vacuous even though only two periodic masks per scale are used:

Proposition 5.4 (Cyclic boundaries and operator-span dimension for periodic Walsh masks)**.** Lean: `walshDigitMask_in_periodicDyadicSpan` [`SwapCodeAnswersProofs.lean`]

Lean: `walshBitSet_cyclicBoundaryCount`, `isCyclicInterval_cyclicBoundaryCount_le_two`, `isPeriodicScaleDyadicMask_cyclicBoundaryCount`, `walshBitSet_joint_separates_points`, `walshDigitMaskRotSpan_finrank_le`, `bonus2_WalshDichotomyStructuralProperties` [`DualityAndSwapConsequences.lean`]

Let $m \ge 0$ *and* $d = 2^m$*.*

(a) Every cyclic interval $I \subseteq \mathrm{Fin}\, d$ *(*$d \ge 1$*) satisfies* $\texttt{cyclicBoundaryCount}(I) \le 2$*, whereas for each* $j \in \mathrm{Fin}\, m$*, both* $W_j = \texttt{walshBitSet}(m, j)$ *and every periodic scale-*j *mask (*`IsPeriodicScaleDyadicMask`*) have* $\texttt{cyclicBoundaryCount} = 2^{j+1}$ *(at least* 4 *once* $j \ge 1$*).*

(b) Across $j \in \mathrm{Fin}\, m$*, the* m *Walsh bit sets* $W_0, \dots, W_{m-1}$ *jointly separate* $\mathrm{Fin}(2^m)$ *into* 2^m *singleton atoms.*

(c) Over every field F*, every family in the span of* $\{\texttt{walshDigitMaskRotMatrix}(F, j, \mathrm{bit}, s) : j \in \mathrm{Fin}\, m,\ \mathrm{bit} \in \{0,1\},\ s \in \mathrm{Fin}(2^m)\}$ *belongs to* `InMultiscalePeriodicDyadicRotSpan` *with constant profile* $M(j) = 2$ *per scale (*$\sum_{j=0}^{m-1} M(j) = 2m = 2\log_2 d$*), and the entire Walsh-masked rotation span has dimension at most* $(m+1)2^m$ *over* F*.*

Outline of the formal proof. In (a), W_j alternates blocks of 2^{m-1-j} zeros and 2^{m-1-j} ones 2^j times around $\mathrm{Fin}(2^m)$, producing 2^{j+1} boundary transitions, and any periodic scale-j mask is a cyclic translate of W_j or its complement. Part (b) recovers $r \in \mathrm{Fin}(2^m)$ from its m binary digits. In (c), the two bit values bit $\in \{0,1\}$ at scale j give two periodic scale-j masks ($M(j) = 2$), while $P_{W_j,0}\sigma^s + P_{W_j,1}\sigma^s = \sigma^s$ shows that the span is generated by the 2^m pure rotations together with the $m \cdot 2^m$ operators $P_{W_j,1}\sigma^s$. □

Whether $\dim_F \texttt{SwapRigidity.swapCode}(A, B)$ is bounded away from d^2 when both A and B lie in the $(m+1)2^m$-dimensional Walsh-masked rotation span remains open.

6 Restricted shuffle-and-local operator pipelines and anti-diagonal untwisting

We state the scope of this section up front. In Section 6.1 we analyze Operator Schmidt Rank ($\mathrm{rk}_\otimes$) across the row/column tensor cut $F^d_{\mathrm{row}} \otimes_F (F^d_{\mathrm{col}} \otimes_F F^b)$ for linear pipelines built from Stone's shuffle and unshuffle [10] together with row-uniform operators $I_d \otimes B$ and column-uniform operators $A \otimes I_{db}$. This invariant applies *only* to that restricted operator pipeline: it *does not* answer [3, Problem 9.5] and *does not* yield a lower bound for Turing machines. Indeed, [3, Problem 9.5] asks for a functional that is subadditive over machine steps, large for the swap at every scale, and *"not raised at all scales by a single linear-time operation"*, whereas within-row cyclic shear has $\mathrm{rk}_\otimes = d$ (Proposition 6.5(a)) yet is computed in $O(L)$ steps on a 2-work-tape Turing machine (Proposition 6.4). Nor do we formalize a converse decomposition of τ into m shuffle passes here; Theorem 6.3 is a pass lower bound for the specified operator class.

6.1 Operator Schmidt Rank on restricted shuffle pipelines and linear-time machines

Write $\texttt{GridState}(F, d, b) := (\mathrm{Fin}\, d \to \mathrm{Fin}\, d \to \mathrm{Fin}\, b \to F)$, separating the row index $r \in \mathrm{Fin}\, d$ from the within-row column-and-lane index $(c, \ell) \in \mathrm{Fin}\, d \times \mathrm{Fin}\, b$. An F-linear operator K on $\texttt{GridState}(F, d, b)$ has *Operator Schmidt Rank at most* R, written $\mathrm{rk}_\otimes(K) \le R$ (`HasOperatorSchmidtRankLe`), if there exist R row matrices $A_i \in \mathrm{Mat}_{d\times d}(F)$ and R within-row linear maps $B_i \in \mathrm{End}_F(F^d \otimes_F F^b)$ such that

$$(KX)_{r,c,\ell} \;=\; \sum_{i=0}^{R-1} \sum_{r'=0}^{d-1} (A_i)_{r,r'} \, \big(B_i(X_{r'})\big)_{c,\ell},$$

and we write $\mathrm{rk}_\otimes(K) = R$ (`HasExactOperatorSchmidtRank`) when $\mathrm{rk}_\otimes(K) \le R$ and no $R' < R$ suffices.

Lemma 6.1 (Submultiplicativity and unit-rank operators)**.** Lean: `operatorSchmidtRankSubmultiplicative` [`OperatorDualityProofs.lean`]
Lean: `isRowOrColUniform_rankLe_one` [`DualityAndSwapConsequences.lean`]
Over any field F:

(a) If $\mathrm{rk}_\otimes(K_1) \le R_1$ and $\mathrm{rk}_\otimes(K_2) \le R_2$, then $\mathrm{rk}_\otimes(K_2 \circ K_1) \le R_2 R_1$.

(b) Every row-uniform column operator $I_d \otimes B$ (`IsRowUniformColOp`, $(KX)_r = B(X_r)$) and every column-uniform row operator $A \otimes I_{db}$ (`IsColUniformRowOp`, $(KX)_{r,c,\ell} = \sum_{r'} A_{r,r'} X_{r',c,\ell}$) satisfies $\mathrm{rk}_\otimes(K) \le 1$.

Outline of the formal proof. Expanding $K_2 \circ K_1$ over $(i_2, i_1) \in \mathrm{Fin}\, R_2 \times \mathrm{Fin}\, R_1$ multiplies the row matrices $A_{2,i_2} A_{1,i_1}$ and composes the within-row maps $B_{2,i_2} \circ B_{1,i_1}$ using $R_2 R_1$ terms; $I_d \otimes B$ and $A \otimes I_{db}$ are single elementary tensors. □

Theorem 6.2 (Operator Schmidt Rank of transpose and Stone's shuffle)**.** Lean: `operatorSchmidtRankSwap`, `operatorSchmidtRankStoneShuffle` [`OperatorDualityProofs.lean`]

Let F be any field and let $b \geq 1$.

(a) For every $d \geq 1$, the matrix transpose $\tau =$ `gridSwapLinear`(F, d, b) *satisfies* $\mathrm{rk}_{\otimes}(\tau) = d^2$.

(b) On $d = 2^m$, Stone's 1-bit cyclic shuffle $S =$ `stoneShuffleGridLinear`(F, m, b) *and unshuffle $S^{-1} =$* `stoneUnshuffleGridLinear`(F, m, b) *are mutual inverses with $\mathrm{rk}_{\otimes}(S) \leq 4$ and $\mathrm{rk}_{\otimes}(S^{-1}) \leq 4$ for all $m \geq 0$, and for every $m \geq 1$ ($d = 2^m \geq 2$),*

$$\mathrm{rk}_{\otimes}(S) \;=\; \mathrm{rk}_{\otimes}(S^{-1}) \;=\; 4.$$

Outline of the formal proof. In (a), decomposing τ by source/target row pairs uses d^2 terms, and evaluating any R-term decomposition on row-localized lane-0 basis inputs produces R matrices spanning $\mathrm{Mat}_{d\times d}(F)$, forcing $R \geq d^2$. In (b), for $m \geq 1$, Stone's shuffle sends (r, c) to (r', c') by moving the top bit of r into the low bit of c' and the top bit of c into the low bit of r'; splitting on (low bit of c', low bit of r') $\in \{0,1\}^2$ gives a 4-term decomposition whose four non-zero quadrant blocks are linearly independent once $2^m \geq 2$ (and symmetrically for S^{-1}). □

Theorem 6.3 (Pass lower bounds for restricted shuffle-and-local pipelines)**.** Lean: `stoneShufflePipelinePassLowerBound` [`OperatorDualityProofs.lean`]

Lean: `boundedOSRPipelinePassLowerBound` [`DualityAndSwapConsequences.lean`]

Let F be any field and let $b \geq 1$.

(a) On $d = 2^m$ ($m \geq 0$), if an operator list $K_1, \ldots, K_n$ has composition `composeGridOps` *equal to τ, where t indices are equal to S or S^{-1} and every remaining operator satisfies* `IsRowUniformColOp` *or* `IsColUniformRowOp` *(of the form $I_d \otimes B$ or $A \otimes I_{db}$), then $(2^m)^2 \leq 4^t$ and $t \geq m$.*

(b) More generally, for any $d \geq 1$ and $R \geq 1$, if a composed pipeline equals τ with t operators satisfying $\mathrm{rk}_{\otimes} \leq R$ and all remaining operators satisfying $\mathrm{rk}_{\otimes} \leq 1$, then $d^2 \leq R^t$, and for $R \geq 2$ one has $t \geq \dfrac{2 \log d}{\log R}$.

Outline of the formal proof. By Lemma 6.1 and Theorem 6.2(b), the composed pipeline has $\mathrm{rk}_{\otimes} \leq 4^t$ (respectively $\mathrm{rk}_{\otimes} \leq R^t$), whereas $\mathrm{rk}_{\otimes}(\tau) = d^2$ by Theorem 6.2(a). Taking real logarithms when $R \geq 2$ gives $2 \log d \leq t \log R$. □

In the shared library `lean-machine` (`LeanMachine.ShuffleShear`), we construct explicit transition tables for one Stone shuffle pass, one Stone unshuffle pass, and within-row cyclic shear, and transport them into `Transpose.TM` in `ShuffleShearBridge.lean`:

Proposition 6.4 (Verified $O(L)$-time multitape Turing machines for shuffle, unshuffle, and row shear)**.** Lean: `stoneShuffleTM_threeTape_linear`, `stoneUnshuffleTM_twoTape_linear`, `rowShearTM_twoTape_linear` [`ShuffleShearBridge.lean`]

With $b =$ entryBits(d) *(*`Transpose.entryBits`*) and $L =$* inputLength$(d) = d^2 b$ *(*`Transpose.inputLength`*):*

(a) `stoneShuffleTransposeTM` *has 3 work tapes ($k = 3$) and, for every $m \geq 1$ ($d = 2^m$) and every matrix A (*`Transpose.Matrix`*), computes* `stoneShuffleTMMatrix` *on (m, A) within $150L$ steps.*

(b) `stoneUnshuffleTransposeTM` *has* 2 *work tapes* ($k = 2$) *and, for every* $m \geq 1$ ($d = 2^m$) *and every matrix* A (`Transpose.Matrix`)*, computes* `stoneUnshuffleTMMatrix` *on* (m, A) *within* $150L$ *steps.*

(c) `rowShearTransposeTM` *has* 2 *work tapes* ($k = 2$) *and, for every* $d \geq 1$ *and every matrix* A (`Transpose.Matrix`)*, computes within-row cyclic shear* `rowShearTMMatrix` *on* (d, A) *(sending entry* (r, c, ℓ) *to* $A_{r,(c+r) \bmod d,\ell}$*) within* $200L$ *steps.*

Outline of the formal proof. Because d and b are not hardcoded in the finite control, the machines in `LeanMachine.ShuffleShear` first determine unary counters for b and d (or $H = d^2/2$) from the input length $L = d^2 b$ in $O(L)$ steps. The 3-work-tape shuffle machine stores unary counters for b and H on two work tapes, buffers the first H words on a third work tape, and interleaves b-bit words from the buffer and the second half of the input tape. The 2-work-tape unshuffle machine emits even-indexed b-bit words directly from the input tape while buffering odd-indexed words on tape 0 using a unary b-counter on tape 1, then replays tape 0. The 2-work-tape row-shear machine copies each db-bit row to tape 0 and emits it cyclically shifted by rb bits. `computesWithin_toTransposeTM` transports each proof identically to `Transpose.ComputesWithin`. □

Two controls clarify the boundary of $\mathrm{rk}_\otimes$ and of single-cell information measures:

Proposition 6.5 (Within-row shear $\mathrm{rk}_\otimes = d$ and row-diagonal source-arity gauge invariance)**.** Lean: `control_RowShearAndRowDiagonalAlternation` [`OperatorDualityProofs.lean`]
Lean: `rowDiagonal_SourceArityGaugeInvariance` [`DualityAndSwapConsequences.lean`]

Let F *be any field,* $d \geq 1$*, and* $b \geq 1$*.*

(a) Within-row cyclic shear `rowShearLinear` *on* (F, d, b) *lies in the row-diagonal subalgebra* `IsRowDiagonalOp` *(*$(KX)_r = B_r(X_r)$*) yet has operator Schmidt rank* d*.*

(b) Output-row source arity (`HasOutputRowSourceArityLe`*) is invariant under two-sided composition* $D_1 \circ K \circ D_0$ *with arbitrary row-diagonal operators* D_0, D_1*.*

(c) On $d = 2^m$ ($m \geq 0$)*, every alternating product* $D_s U_s D_{s-1} \cdots U_1 D_0$ (`alternatingRowDiagProduct`) *of* $s+1$ *row-diagonal operators* D_i *and* s *shuffle or unshuffle passes* $U_j \in \{S, S^{-1}\}$ *has output-row source arity at most* 2^s*; if it equals* τ*, then* $2^m \leq 2^s$ *and* $s \geq m$*.*

Outline of the formal proof. In (a), `rowShearLinear` $= \sum_{r=0}^{d-1} E_{r,r} \otimes \sigma^r$, and linear independence of the d cyclic column shifts $\sigma^0, \ldots, \sigma^{d-1}$ forces $\mathrm{rk}_\otimes = d$. In (b) and (c), row-diagonal operators act within each row without mixing source rows, while each S or S^{-1} combines at most two source rows into each output row; since row 0 of $\tau(X)$ depends on all 2^m input rows when $b \geq 1$, equality with τ forces $2^m \leq 2^s$. □

Proposition 6.6 (Skew one-time-pad zero-mutual-information control)**.** Lean: `control_SkewOneTimePadInformationTrap` [`OperatorDualityProofs.lean`]
Lean: `uniformShannonMI_eq_zero_of_indep`, `skewOneTimePad_ShannonMutualInfoZero` [`DualityAndSwapConsequences.lean`]

For $d \geq 1$ *and* $b \geq 1$*, define* $W(X) := X^{\mathsf{T}} \oplus X$ *on* $\{0,1\}^{d \times d \times b}$ (`skewXorPad`)*. Then* $W(X) \oplus X = X^{\mathsf{T}}$ *for every* X*, and under uniformly random* X*, every single entry* $W_{r,c}$ *is statistically independent of every target column* $X_{:,c'}$*, so its discrete Shannon mutual information (*`uniformShannonMI`*) satisfies*

$$I(W_{r,c}\,;\, X_{:,c'}) \;=\; 0 \qquad \text{for all } r, c, c' \in \mathbb{Z}_d.$$

Outline of the formal proof. When $r = c$, $W_{r,r} = 0$ is constant. When $r \neq c$, the two entries $X_{c,r}$ and $X_{r,c}$ lie in distinct columns $r \neq c$, so at least one lies outside column c' and acts as an independent uniform pad on the other, giving exact product-count independence and zero Shannon mutual information. □

6.2 Anti-diagonal untwisting of shift-collision equations

We close this section with a coordinate observation on linear codes with deterministic shared decoders (`SharedDecoders.IsSharedCode`, [6, Section 2]) and the shift-link covariance spaces of [3, Section 6]. Untwisting matrix coordinates along anti-diagonals $a = c + j \in \mathbb{Z}_d$ via the mutually inverse F-linear maps (`antiDiagLinear`, `fromAntiDiagLinear`)

$$Z_{c,a} \;:=\; X_{c,\,a-c}, \qquad X_{c,j} \;=\; Z_{c,\,j+c} \qquad (c, a, j \in \mathbb{Z}_d)$$

induces for $d \geq 1$ the linear tensor isomorphism $\Psi = \texttt{antiDiagTensorEquiv} : (\mathbb{Z}_d \to F) \otimes_F (\mathbb{Z}_d \to A) \xrightarrow{\sim} \mathrm{Mat}_{d\times d}(A)$ sending $u \otimes v$ to $(c, j) \mapsto u_c\, v_{j+c}$.

Theorem 6.7 (Pairwise shift-collision characterization and anti-diagonal tensor separation)**.** Lean: `pairwiseShiftCollisionCharacterization`, `antiDiagonalTensorSeparation` [`OperatorDualityProofs.lean`]

Lean: `antiDiagTensorEquiv_intertwines_rot`, `shiftSetCovariance_tensor_finrank` [`DualityAndSwapConsequences.lean`]

Let F be a field, $d \geq 1$, A an F-vector space, and $p : \mathbb{Z}_d \to \kappa$.

(a) A linear subspace $C \leq \mathrm{Mat}_{d\times d}(A)$ satisfies `SharedDecoders.IsSharedCode` *for p if and only if for every shift $s \in \mathbb{Z}_d$ and every $Y \in C$ with $\mathrm{rot}_s(Y) \in C$, Y lies in* `InPairwiseShiftCovariance`*: whenever $p(c) = p(c + s)$, one has $Y_{c+s,\,j} = Y_{c,\,j+s}$ for all $j \in \mathbb{Z}_d$.*

(b) Under $\Psi : (\mathbb{Z}_d \to F) \otimes_F (\mathbb{Z}_d \to A) \xrightarrow{\sim} \mathrm{Mat}_{d\times d}(A)$, column rotation rot_t intertwines with the pure right-factor coordinate shift $\mathrm{id} \otimes \sigma^t$ (`coordShiftLinear`*, $(\sigma^t v)_a = v_{a+t}$).*

(c) For any shift set $E \subseteq \mathbb{Z}_d$, a matrix $X \in \mathrm{Mat}_{d\times d}(A)$ lies in the shift-set covariance subspace $H_E(p)$ (`shiftSetCovarianceSubmodule`*) if and only if its anti-diagonal coordinates $Z_{c,a} = X_{c,a-c}$ are constant in c on every connected component of* `SwapRigidity.Link` *p E (*`antiDiagonalTensorSeparation`*). When $\dim_F A < \infty$, writing $U_E(p) \leq (\mathbb{Z}_d \to F)$ for the subspace of vectors constant on those components (*`leftComponentSubmodule`*), Ψ maps $U_E(p) \otimes_F (\mathbb{Z}_d \to A)$ onto $H_E(p)$ and*

$$\dim_F H_E(p) \;=\; \dim_F U_E(p) \cdot d \cdot \dim_F A.$$

Outline of the formal proof. In (a), determinism of $f_{p(c)}$ on a linear subspace C means that whenever $X_1, X_2 \in C$ satisfy $p(c_1) = p(c_2)$ and $\mathrm{rot}_{c_1}(X_1) = \mathrm{rot}_{c_2}(X_2)$, row c_1 of X_1 equals row c_2 of X_2; setting $s = c_2 - c_1$ and $Y = X_1 - \mathrm{rot}_{-s}(X_2) \in C$ gives $\mathrm{rot}_s(Y) = \mathrm{rot}_s(X_1) - X_2 \in C$ and the pairwise identity. In (b), $\Psi(\mathrm{id} \otimes \sigma^t(u \otimes v))_{c,j} = u_c v_{j+c+t} = \mathrm{rot}_t(\Psi(u \otimes v))_{c,j}$. In (c), substituting $X_{c,j} = Z_{c,j+c}$ turns $X_{c',j+\delta'} = X_{c,j+\delta}$ under $c + \delta = c' + \delta'$ into $Z_{c',a} = Z_{c,a}$ independently for each anti-diagonal $a = c + \delta + j$. □

Proposition 6.8 (Two-subspace Grassmann relaxation and alternating-mask shift separation)**.**
Lean: `maskFloor_GenericSubspaceRealization` [`DualityAndSwapConsequences.lean`]

Let F be a field, $d \geq 1$, and A a finite-dimensional F-vector space.

(a) For any $p : \mathbb{Z}_d \to \kappa$, any shift $s \in \mathbb{Z}_d$, and any subspaces $C_0, C_s \leq \mathrm{Mat}_{d\times d}(A)$ with $\dim_F C_0 = \dim_F C_s$ and $C_0 \cap C_s \leq H_{\{0,s\}}(p)$, Grassmann modularity gives $2\dim_F C_0 \leq d^2 \dim_F A + \dim_F H_{\{0,s\}}(p)$. Conversely, once the right-factor shift coupling $C_s = \mathrm{rot}_{-s}(C_0)$ is dropped, splitting a complement of $U_{\{0,s\}}(p)$ in $(\mathbb{Z}_d \to F)$ into two equal-sized subspaces constructs $C_0 = \Psi(V_0 \otimes \top)$ and $C_s = \Psi(V_s \otimes \top)$ with $C_0 \cap C_s = H_{\{0,s\}}(p)$, $\dim_F C_0 = \dim_F C_s$, and

$$d^2 \dim_F A + \dim_F H_{\{0,s\}}(p) \;\leq\; 2\dim_F C_0 + d \dim_F A.$$

(b) On even $d = 2m$ ($m \geq 2$) with alternating partition $p(c) = (c \bmod 2 == 1)$, shift $s = 1$ satisfies $p(c) \neq p(c+1)$ everywhere so $U_{\{0,1\}}(p) = \top$ and $H_{\{0,1\}}(p) = \top$, whereas shift $s = 2$ has $\dim_F U_{\{0,2\}}(p) = 2$ and forces $2\dim_F C \leq ((2m)^2 + 4m)\dim_F A$ for every shared code C.

Outline of the formal proof. Part (a) applies $\dim(C_0 + C_s) + \dim(C_0 \cap C_s) = 2\dim C_0$ for the upper bound, and chooses disjoint sets of $\lfloor (d - \dim U_{\{0,s\}}(p))/2 \rfloor$ non-representative indices on the left tensor factor for the converse construction. Part (b) checks that shift 1 has no monochromatic links on the alternating coloring while shift 2 connects all even indices and all odd indices into two components. □

7 Cut-cone evaluation, random-order capacity, and per-entry cut labels

7.1 Cut-causal cone evaluation on frozen `Transpose.TM` fibres

Let M be a deterministic multitape Turing machine (`Transpose.TM`) with $1 \leq M.k$ (available without loss of generality for any transposer by Lemma 2.2), and let $C \subseteq \Sigma_d^{d\times d}$ be a frozen-trace Γ-cut fibre (`IsFrozenTraceCutFibre`) over a step horizon T: all $X \in C$ halt at step T with output X^{T}, follow a common input/work head trajectory (inPos, workPos) and output emission schedule emitStep, and agree on the post-step control state and written work-tape symbols at every cut step $u \in \Gamma \subseteq \{0, \ldots, T-1\}$. Along workPos, a direct causal step $u \to v$ (`TMDepStep`) holds for $u < v < T$ when either $v = u + 1$ (the control-state spine edge) or u is the most recent step visiting the cell scanned by one of the $M.k$ work heads at step v (`IsLastWorkVisit`). A descending path is Γ-free (`IsCutFreeCausalPath`) if none of its strict predecessors lies in Γ; the resulting Γ-truncated causal cone of step v is `trueConeSteps` (`InCutCausalCone`), and its syntactic input-cell support is `cutConeInputCells`.

In `MachineSweepConsequences.lean`, we also represent each cone syntactically as a rooted $(1 + M.k)$-ary symbolic tree (`CausalConeTree`) extracted relative to column c (`traceConeTree`) and evaluated bottom-up on $\mathrm{rot}_c(X)$ directly through $M.\delta$ (`evalConeTree`, syntactic support `treeCellSupport`):

Lemma 7.1 (Syntactic cut-cone support and symbolic tree evaluation on frozen fibres)**.** Lean: `tmTraceCutConeDetermination` [`NativeSweepInterfaceProofs.lean`]

Lean: `evalConeTree_cellSupport`, `evalConeTree_entryCellSupport_card_le`, `evalConeTree_traceConeTree_eq` [`MachineSweepConsequences.lean`]

Let M be a `Transpose.TM` *with $1 \le M.k$, let $d \ge 1$ and $D \ge 1$, and let $C \subseteq \Sigma_d^{d\times d}$ satisfy* `IsFrozenTraceCutFibre` *such that every Γ-free causal path has at most D vertices (π.length $\le D$).*

(a) For every step $v < T$, $|\texttt{trueConeSteps}(\mathrm{workPos}, T, \Gamma, v)| \le (M.k+1)^D$. For every output bit (r, c, ℓ) emitted at step $v = \mathrm{emitStep}(\texttt{outputBitIdx}(d, r, c, \ell))$, the explicit syntactic cone support $Q_{\mathrm{bit}} := \texttt{cutConeInputCells}(d, T, \mathrm{inPos}, \mathrm{workPos}, \Gamma, v)$ consists of those cells visited by inPos *on* `trueConeSteps`*, satisfies $|Q_{\mathrm{bit}}| \le (M.k+1)^D$, and uniquely determines $X_{c,r,\ell}$ across C: whenever $X_1, X_2 \in C$ agree on Q_{bit}, one has $X_1(c, r, \ell) = X_2(c, r, \ell)$.*

(b) Every depth-D `CausalConeTree` *has $|\texttt{treeCellSupport}(\mathrm{tree})| \le (M.k+1)^D$ and depends on $\mathrm{rot}_c(X)$ only through* `treeCellSupport`(tree)*. Given any cut-value assignment* cutVal *agreeing with the frozen Γ-cut states and work symbols on C, evaluating* `traceConeTree` *at depth D on $\mathrm{rot}_c(X)$ ($X \in C$) via* `evalConeTree` *reproduces the post-step control state, written work symbols, and emitted output bit $X_{c,r,\ell}$ of the live machine run. Across all $b = \texttt{Transpose.entryBits}(d)$ bit-lane trees of an output entry, the union of* `treeCellSupport` *has size at most $\Delta = b(M.k+1)^D$.*

Outline of the formal proof. Each step $w < T$ has at most $1 + M.k$ direct causal predecessors ($w-1$ for the control state and at most one last-visit step per work tape), so inducting on the depth budget $m \le D$ gives the geometric sum bound $1 + (M.k+1)((M.k+1)^{m-1} - 1) < (M.k+1)^m$. Defining $Q_{\mathrm{bit}} =$ `cutConeInputCells` explicitly as the union of input cells read along `trueConeSteps` avoids self-referential singleton footprints: strong induction on $w < T$ shows that inputs in C agreeing on Q_{bit} induce identical transitions $M.\delta$ throughout the Γ-stopped cone of v, and `evalConeTree` replays those exact transitions bottom-up from initial blank leaves and frozen Γ-cut leaves on $\mathrm{rot}_c(X)$. □

7.2 Lean formalization of the $1/(\Delta+1)$ random-order bound for shared set codes

For any recovery relation of maximum footprint size Δ, the acyclic-subgraph bound of Bar-Yossef, Birk, Jayram, and Kol [1, Theorem 3] combined with a uniformly random vertex ordering gives a maximum acyclic induced subgraph of size at least $(\#\mathrm{clean})/(\Delta+1)$ and hence rate at most $1 - 1/(\Delta+1)$ up to self-referential entries. This elementary bound has been in the series since [4, Proposition 4.14(a)–(b)] (and is summarized in [3, Sections 4.1 and 6.7]). We record its Lean 4 formalization in the notation of Definition 2.1.

Given $p : \mathbb{Z}_d \to \mathrm{Fin}\, M$ and $Q : \mathrm{Fin}\, M \to \mathbb{Z}_d \to \mathrm{Finset}(\mathbb{Z}_d \times \mathbb{Z}_d)$, evaluating $f_{p(c)}(\mathrm{rot}_c(X))_r$ reads the shifted footprint (`shiftedFootprint`)

$$F_{c,r} \;:=\; \{(r',\, c+u) \;:\; (r', u) \in Q_{p(c),r}\} \subseteq \mathbb{Z}_d \times \mathbb{Z}_d,$$

and an input entry (c, r) is *clean* (`cleanEntries`) if $(r, c) \notin \mathcal{E}$ and $(c, r) \notin F_{c,r}$. Note that decoder sharing enters the following bound *only* through the $dM\Delta$ count of self-referential entries $(c, r) \in F_{c,r}$; once clean entries are isolated, the random-order bound uses only the individual footprints $F_{c,r}$.

Lemma 7.2 (Self-referential entries and clean-entry count)**.** Lean: `mem_shiftedFootprint_self_iff`, `selfReferentialAndCleanCount` [`BoundedAritySetCapacityProofs.lean`]

For all $d \geq 1$, $M, \Delta \in \mathbb{N}$, $p : \mathbb{Z}_d \to \operatorname{Fin} M$, $Q : \operatorname{Fin} M \to \mathbb{Z}_d \to \operatorname{Finset}(\mathbb{Z}_d \times \mathbb{Z}_d)$, and $\mathcal{E} \subseteq \mathbb{Z}_d \times \mathbb{Z}_d$:

(a) $(c, r) \in F_{c,r}$ if and only if $(c, r - c) \in Q_{p(c),r}$.

(b) If $|Q_{m,r}| \leq \Delta$ for all (m, r), then $|F_{c,r}| \leq \Delta$ for every clean entry (c, r) and

$$|\texttt{cleanEntries}(p, Q, \mathcal{E})| \;\geq\; d^2 - |\mathcal{E}| - dM\Delta.$$

Outline of the formal proof. For each fixed (m, r), the map $c \mapsto (c, r - c)$ is injective on $\mathbb{Z}_d$, so at most $|Q_{m,r}| \leq \Delta$ columns $c \in p^{-1}(m)$ satisfy $(c, r) \in F_{c,r}$. Summing over $m \in \operatorname{Fin} M$ and $r \in \mathbb{Z}_d$ bounds self-referential entries by $dM\Delta$. □

Theorem 7.3 (Random-order capacity bound for bounded-arity shared set codes; after [4, Proposition 4.14(a)–(b)] and [1, Theorem 3])**.** Lean: `orderedRecoverable_injective_and_card_le`, `exists_orderedRecoverableCells_bound`, `randomOrder_BoundedAritySetCapacity`, `corollary_OneClassArityOneHalf`, `corollary_OneClassArityTwoTwoThirds`, `asymptotic_SublinearClassesBrokenFractionRefutesRateOne` [`SetCapacityConsequences.lean`]

Let $d \geq 1$, let $M, \Delta \in \mathbb{N}$, let Σ be a finite alphabet with $|\Sigma| \geq 2$, and let $C \subseteq \Sigma^{d \times d}$ satisfy $\texttt{IsBoundedAritySetCode}(\Delta, p, \mathcal{E}, f, Q, C)$.

(a) For every ranking $\mathrm{rk} : \mathbb{Z}_d \times \mathbb{Z}_d \to \mathbb{N}$, let $A(\mathrm{rk}) := \texttt{orderedRecoverableCells}(p, Q, \mathcal{E}, \mathrm{rk})$ be the set of clean entries (c, r) satisfying $\mathrm{rk}(y) < \mathrm{rk}(c, r)$ for all $y \in F_{c,r}$. Then restriction to $(\mathbb{Z}_d \times \mathbb{Z}_d) \setminus A(\mathrm{rk})$ is injective on C, so $|C| \leq |\Sigma|^{d^2 - |A(\mathrm{rk})|}$, and averaging over all permutations of $\mathbb{Z}_d \times \mathbb{Z}_d$ produces an injective ranking rk_* with $(\Delta + 1)|A(\mathrm{rk}_*)| \geq |\texttt{cleanEntries}(p, Q, \mathcal{E})|$.

(b) Consequently, for every (empty or nonempty) code C,

$$\frac{\log|C|}{\log|\Sigma|} \;\leq\; d^2 \;-\; \frac{d^2 - |\mathcal{E}| - dM\Delta}{\Delta + 1}.$$

At $M = 1, \mathcal{E} = \emptyset$, this gives rate at most $\frac{1}{2} + \frac{1}{2d}$ at $\Delta = 1$ and $\frac{2}{3} + \frac{2}{3d}$ at $\Delta = 2$; and for any fixed $\Delta \geq 1$, $\delta \in (0, 1]$, $|\mathcal{E}_d| \leq (1 - \delta)d^2$, and $M_d = o(d/\Delta)$, the rate is eventually at most $1 - \frac{\delta}{2(\Delta+1)}$.

Outline of the formal proof. In (a), strong induction on $\mathrm{rk}(c, r)$ shows that two codewords agreeing outside $A(\mathrm{rk})$ agree on $A(\mathrm{rk})$ as well, because every $y \in F_{c,r}$ strictly precedes (c, r). For each clean entry $x = (c, r)$, $|F_{c,r} \cup \{x\}| \leq \Delta + 1$; swapping x with the τ-maximum element of $F_{c,r} \cup \{x\}$ partitions the symmetric group so that x comes last in $F_{c,r} \cup \{x\}$ on at least a $1/(\Delta + 1)$ fraction of permutations, and double counting selects rk_*. Combining (a) with Lemma 7.2(b) and $\log|\Sigma| > 0$ gives (b) (with $\log 0 = 0$ in Lean handling $C = \emptyset$). □

Remark 7.4 (Comparison with small-window and unrestricted-arity bounds)**.** In `BoundedAritySetCapacityProofs.lean` (`oneHopRecoverableInjective`, `existsRecoverableSubsetBound`, `theorem1_BoundedAritySetCapacity`, `growingClassesPositiveDeficit`) we also formalized the weaker one-hop product-Bernoulli variant with factor $\Delta^\Delta/(\Delta+1)^{\Delta+1} \in (1/(e(\Delta+1)), 1/(2(\Delta+1))]$, together with boundary constructions at $\Delta = d$, $\Delta = 2d - 1$, and $\Delta = d^2$ (`boundaryControl_DeficitVanishesAtDeltaEqD`, `unrestrictedArity_RateLowerBoundAndTendstoOne`, `unrestrictedArity_ExactLogarithmicAsymptoticAllDimensions`, wrapping the $|C| \geq |\Sigma|^{d^2}/(2d)$ orbit

construction of [6, Theorem A.1 and Corollary A.2], where full-view nonlinear codes reach rate $1 - o(1)$). At $\Delta \le 2$, stronger results (proved in text, allowing a separate recovery function for each position) are already in [3]: at window $w = 1$, [3, Proposition 6.21(b)] gives the same exponent $|C| \le q^{(d^2+Md)/2}$ (rate $\frac{1}{2} + \frac{1}{2d}$ at $M = 1$) without requiring shared functions, and at window $w \le 2$, [3, Proposition 6.22(b)] improves on the $\frac{2}{3} + \frac{2}{3d}$ bound of Theorem 7.3(b) to $|C| \le q^{d(d+1)/2+d-1}$ (rate $\frac{1}{2} + O(1/d)$). Meanwhile, [3, Problem 9.4(b)] asks whether rate $\frac{1}{2} + o(1)$ holds at fixed windows $w \ge 3$ over a *fixed* alphabet of q symbols, rather than merely rate < 1.

7.3 Why column classes cannot absorb per-entry cut labels

Remark 7.5 (Two obstructions to bridging periodic stretches via column-class set capacity)**.** It is natural to ask whether Lemma 7.1 and Theorem 7.3 can bound a periodic stretch of a Turing machine by conditioning on a bounded-depth trace cut Γ. Two separate obstructions prevent this:

(1) *Depth-D cut cost exceeds the $1/(\Delta+1)$ deficit ([4, Propositions 4.13 and 4.14(c)]).* Cutting all causal paths down to length D over a run of length T requires revealing a cut word of bit-cost $\varepsilon L = \Omega(L/D)$, whereas $(k+1)^{D+1} > 1/\varepsilon$ and the entry-level cone arity $\Delta = b(M.k+1)^D$ grows exponentially in D (and includes the word-width factor $b = \lceil \log_2 d \rceil$). Thus an $O(1/\Delta)$ rate deficit is always smaller than the cut-advice rate and, as already proved in [4, Proposition 4.14(c)], cannot by itself contradict a transposer.

(2) *Column classes cannot absorb per-entry cut labels.* Even before comparing $1/(\Delta+1)$ against cut advice, the numerator $d^2 - |\mathcal{E}| - dM\Delta$ in Theorem 7.3(b) is *strictly negative* whenever one attempts to turn a Γ-cut fibre of depth D into a column-class code, either by discarding output entries whose cones touch Γ or by refining column classes using L_{cut} frozen cut slots per column (Proposition 7.6 below). Because `TMDepStep` includes the control-state spine edge $u \to u+1$, every emission step $v \ge D$ encounters a cut step $u \in \Gamma \cap [v-D, v)$ along the spine alone. Consequently, cut labels are attached to individual output entries rather than whole columns, which is why bridging periodic machine stretches requires the *per-position* fixed-alphabet formulation of [3, Problem 9.4(b)] rather than column-shared decoder classes.

Proposition 7.6 (Negative brackets for per-column cut absorption)**.** `Lean`: `perColumnCutAbsorption_bracket_neg` [`MachineSweepConsequences.lean`]

Let M be a `Transpose.TM` *with $1 \le M.k$, let $d \ge 1$ and $D \in \mathbb{N}$, and let $C \subseteq \Sigma_d^{d\times d}$ be a* nonempty *fibre satisfying* `IsFrozenTraceCutFibre` *whose Γ-free causal paths have at most D vertices. Set $\Delta := $* `Transpose.entryBits`$(d) \cdot (M.k+1)^D$.

(a) Discarding cut-touching outputs*: let $\mathcal{E}_{\mathrm{cut}}$ (*`cutTouchingOutputCells` *for d, T,* workPos*,* emitStep*, and Γ) be the set of output coordinates (r,c) having a bit cone that meets Γ along a direct* `TMDepStep`*. Then $d^2 \le |\mathcal{E}_{\mathrm{cut}}| + D$, and for every $M_0 \ge 1$ and every $\mathcal{E}_{\mathrm{sched}} \subseteq \mathbb{Z}_d \times \mathbb{Z}_d$, setting $\mathcal{E} := \mathcal{E}_{\mathrm{sched}} \cup \mathcal{E}_{\mathrm{cut}}$ gives*

$$d^2 - |\mathcal{E}| - dM_0\Delta \;<\; 0.$$

(b) Refining column classes by per-column cut slots*: suppose every cut step $u \in \Gamma$ touched by a bit cone of an unbroken output $(r,c) \notin \mathcal{E}_{\mathrm{sched}}$ lies among L_{cut} column-indexed cut slots* $\mathrm{colCutStep}(c) : \mathrm{Fin}\, L_{\mathrm{cut}} \to \mathbb{N}$*. Then $d^2 \le |\mathcal{E}_{\mathrm{sched}}| + D + dL_{\mathrm{cut}}D$, and for every $M_0 \ge 1$, whenever $(M.q+1)(M.g+1)^{M.k} \ge 2$, the refined class count $M_{\mathrm{cls}} := M_0\big((M.q+1)(M.g+1)^{M.k}\big)^{L_{\mathrm{cut}}}$ satisfies*

$$d^2 - |\mathcal{E}_{\mathrm{sched}}| - dM_{\mathrm{cls}}\Delta \;<\; 0.$$

Outline of the formal proof. Since $C \neq \emptyset$, the d^2 lane-0 emission steps $v(r,c) = \mathrm{emitStep}(\mathrm{idx}_0(r,c))$ (via `outputBitIdx` at lane 0) are pairwise distinct. For any (r,c) with $v(r,c) \geq D$, the descending control-state spine from $v(r,c)$ cannot avoid Γ for D steps without creating a Γ-free path of $D+1$ vertices; taking the largest $u \in \Gamma \cap [v(r,c)-D, v(r,c))$ places $u+1$ in the Γ-cut cone of $v(r,c)$ with $u \to u+1$, so $(r,c) \in \mathcal{E}_{\mathrm{cut}}$. At most D emission steps satisfy $v(r,c) < D$, giving $d^2 - |\mathcal{E}_{\mathrm{cut}}| \leq D < 2^D \leq \Delta \leq dM_0\Delta$ and proving (a). In (b), each slot $u = \mathrm{colCutStep}(c,s)$ can serve as that most recent cut step for at most D emission steps $v(r,c) \in (u, u+D]$, leaving at most $D + dL_{\mathrm{cut}}D \leq d(L_{\mathrm{cut}}+1)D$ unbroken entries; meanwhile $(M.q+1)(M.g+1)^{M.k} \geq 2$ implies $M_{\mathrm{cls}} \geq 2^{L_{\mathrm{cut}}} \geq L_{\mathrm{cut}}+1$, and $D < \Delta$ yields $d(L_{\mathrm{cut}}+1)D < dM_{\mathrm{cls}}\Delta$. □

7.4 Open problems

Question (T) remains open. Within the models studied in this paper and in [3, 6], three concrete algebraic and combinatorial problems remain unresolved:

1. **Macroscopic rank deficit and positive characteristic in two-stage linear composition.** Can the $\Omega(bd(\log d)^{1/(2M_2-1)})$ error-rank bound of Theorem 4.6 be strengthened to $\Omega_{M_1,M_2}(bd^2)$ for fixed M_1, M_2, and extended to fields in which $\mathrm{char}(F) \mid d$ (such as $\mathbb{F}_2$ at $d = 2^m$)?

2. **Remaining dimensions in Remark 5.6 and Walsh-masked swap codes in Problem 9.3.** Does any three-term swap code exceed $d(d+1)/2$ at $d = 5$ or at any $d \geq 8$ outside $\{9, 12\}$ ([3, Remark 5.6])? And on $d = 2^m$, when both A_a and B_a lie in the $(m+1)2^m$-dimensional span of periodic Walsh digit masks composed with rotations (Proposition 5.4), is $\dim_F \texttt{SwapRigidity.swapCode}(A,B) \leq (1-c)d^2$ for a constant $c > 0$ ([3, Problem 9.3])?

3. **Per-position fixed-alphabet windows and machine reductions ([3, Problems 9.4 and 9.5]).** Over a fixed finite alphabet of q symbols and a fixed window $w \geq 3$ on a spread one-class relation, does the $\frac{1}{2} + o(1)$ rate bound hold with a separate recovery function for each position ([3, Problem 9.4(b)]), as needed when cut labels vary from entry to entry (Remark 7.5)? Beyond periodic stretches, finding a machine-level reduction from adaptive schedules or a subadditive multi-scale invariant that is unaffected by single linear-time operations such as within-row shear ([3, Problems 9.4(a) and 9.5]) remains the central barrier to (T).

A Formalization architecture

The Lean 4 formalization is split between the shared repository package `lean-machine` (`../lean-machine`) and this paper's package `BoundedArityDecoders` (`lean/`), both pinned to Lean 4 and Mathlib version `v4.34.1` [11]. Through Lake path dependencies in `lean/lakefile.toml`, `BoundedArityDecoders` requires `SharedDecoders` ([6], which transitively imports `SwapRigidity` from [3] and `Transpose` from [5]) together with `LeanMachine`. None of the earlier papers is modified, and in `lean-machine` only the package root, axiom audit, and `README` are appended to register four new reusable modules under `LeanMachine/ShuffleShear/`:

- `Model.lean`, `StoneShuffle.lean`, `StoneUnshuffle.lean`, and `RowShear.lean`: standalone deterministic multitape Turing machine semantics and verified linear-time machines for Stone's shuffle (3 work tapes, $\leq 150L$ steps), Stone's unshuffle (2 work tapes, $\leq 150L$ steps), and within-row cyclic shear (2 work tapes, $\leq 200L$ steps).

  Inside `lean/BoundedArityDecoders/`, eighteen public modules are organized into three layers:

(1) *Six definition and specification modules* (`AffineCubeComposition.lean`, `SumsetRankDeficit.lean`, `SwapCodeAnswers.lean`, `OperatorDuality.lean`, `NativeSweepInterface.lean`, and `BoundedAritySetCapacity.lean`), defining the algebraic operators, syntactic causal cones, and named proposition types;

(2) *Six primary proof modules* (`AffineCubeCompositionProofs.lean`, `SumsetRankDeficitProofs.lean`, `SwapCodeAnswersProofs.lean`, `OperatorDualityProofs.lean`, `NativeSweepInterfaceProofs.lean`, and `BoundedAritySetCapacityProofs.lean`), proving the core theorems and finite controls; and

(3) *Six consequence and bridge modules* (`AffineCubeConsequences.lean`, `SumsetRankConsequences.lean`, `DualityAndSwapConsequences.lean`, `ShuffleShearBridge.lean`, `MachineSweepConsequences.lean`, and `SetCapacityConsequences.lean`), proving the explicit constants $c(M_1, M_2)$ and $D_0(M_1, M_2)$, the general-field finite controls at $d = 13$, $d = 4$, and $d = 12$, the `TensorProduct` anti-diagonal equivalences, the transport of `LeanMachine.ShuffleShear` to `Transpose.TM`, the symbolic `CausalConeTree` evaluator, the $1/(\Delta + 1)$ random-order bound, and the negative per-column cut-absorption bounds.

Every numbered theorem, lemma, proposition, corollary, and example in the manuscript is proved as a `theorem` declaration in these modules; there are no unformalized or partially formalized numbered results.

B Index of Lean declarations

The following declarations are cited by formal-status lines. Names are relative to `BoundedArityDecoders`. A partial status retains the gap stated at its theorem.

`AffineCubeCompositionProofs.lean` `cauchySchwarzAffineCubeExtraction`, `control_D13TwoCubeAPFreeSharpness`, `control_D4SameIntervalCodedFactorization`, `lemma_AffineCubeCommonMessageGrowth`, `theorem2A_TwoStageRamseyFreeCutoff`.

`AffineCubeConsequences.lean` `control_D13_GeneralFieldAndProgressionBridge`, `control_D4_AllWidthsAndBinaryFields`, `denseSubset_AffineCubeCommonMessageBound`, `singleSubset_CauchySchwarzAffineCubeExtraction`, `theorem2A_GrowingClassesNonFactorization`.

`BoundedAritySetCapacityProofs.lean` `mem_shiftedFootprint_self_iff`, `selfReferentialAndCleanCount`.

`DualityAndSwapConsequences.lean` `antiDiagTensorEquiv_intertwines_rot`, `bonus1_UnifiedThreeTermExceedsPairingBound`, `bonus2_SublinearProfileAndTendstoHalf`, `bonus2_WalshDichotomyStructuralProperties`, `boundedOSRPipelinePassLowerBound`, `isCyclicInterval_cyclicBoundaryCount_le_two`, `isPeriodicScaleDyadicMask_cyclicBoundaryCount`, `isRowOrColUniform_rankLe_one`, `maskFloor_GenericSubspaceRealization`, `rowDiagonal_SourceArityGaugeInvariance`, `shiftSetCovariance_tensor_finrank`, `skewOneTimePad_ShannonMutualInfoZero`, `uniformShannonMI_eq_zero_of_indep`, `walshBitSet_cyclicBoundaryCount`, `walshBitSet_joint_separates_points`, `walshDigitMaskRotSpan_finrank_le`.

`MachineSweepConsequences.lean` `evalConeTree_cellSupport`, `evalConeTree_entryCellSupport_card_le`, `evalConeTree_traceConeTree_eq`, `padDummyWorkTape_transposesWithin`, `perColumnCutAbsorption_bracket_neg`.

NativeSweepInterfaceProofs.lean entryBits_eq_clog, tmTraceCutConeDetermination.

OperatorDualityProofs.lean antiDiagonalTensorSeparation, control_RowShearAndRowDiagonalAlternation, control_SkewOneTimePadInformationTrap, operatorSchmidtRankStoneShuffle, operatorSchmidtRankSubmultiplicative, operatorSchmidtRankSwap, pairwiseShiftCollisionCharacterization, stoneShufflePipelinePassLowerBound.

SetCapacityConsequences.lean asymptotic_SublinearClassesBrokenFractionRefutesRateOne, canonicalWordAlphabet_card_and_ge_two, corollary_OneClassArityOneHalf, corollary_OneClassArityTwoTwoThirds, exists_orderedRecoverableCells_bound, orderedRecoverable_injective_and_card_le, randomOrder_BoundedAritySetCapacity.

ShuffleShearBridge.lean rowShearTM_twoTape_linear, stoneShuffleTM_threeTape_linear, stoneUnshuffleTM_twoTape_linear, tmMatrix_gridLinear_bridges.

SumsetRankConsequences.lean control_D12_BandRowSpaceChannelEquiv, control_D12_UnconditionalExists, d12AddChar_injective_and_primitive, exists_primitive_twelfth_root_complex, finsetNatDiam_shift_min, sumsetAnnihilatingPolynomial_bandRowSpace, theorem2BExplicitConst_pos, theorem2BExplicitThreshold_eq_pow, theorem2B_ExplicitConstant, theorem2B_OneSecondStage_OmegaLog, unshiftedToShiftedSumset_intrinsicDiam.

SumsetRankDeficitProofs.lean control_D12SingleClassAbstractBand, cyclicChannelShift_aeval_ker_finrank_le, iteratedKovariSosTuranSumset, stackedBandPullbackInjective, sumsetAnnihilatingPolynomial, sumsetTwoHopRankBound, theorem2B_TwoStageLogPowerRankDeficit, unshiftedToShiftedSumset, vandermondeArbitrarySubsetInjective.

SwapCodeAnswersProofs.lean bonus1_ThreeTermBlockSwapCodeDim, bonus1_ThreeTermPaddedSevenAndFive, bonus2_MultiscaleFullProfileAttainsFullSwap, bonus2_MultiscaleIntervalBothSides, walshDigitMask_in_periodicDyadicSpan.

Other declarations named in the text CanonicalWordAlphabet, CausalConeTree, GridState, HasExactOperatorSchmidtRank, HasOperatorSchmidtRankLe, HasOutputRowSourceArityLe, InAlignedDyadicIntervalRotSpan, InCutCausalCone, InMultiscaleIntervalRotSpan, InMultiscalePeriodicDyadicRotSpan, InPairwiseShiftCovariance, IsAlignedDyadicInterval, IsBoundedAritySetCode, IsColUniformRowOp, IsCutFreeCausalPath, IsCyclicInterval, IsFrozenTraceCutFibre, IsLastWorkVisit, IsPeriodicScaleDyadicMask, IsRowDiagonalOp, IsRowUniformColOp, IsThreeByThreeSymmetricZeroTraceBlocks, SyntacticCellSupport, SyntacticColumnSupport, TMDepStep, ThreeAPFreeMod, alternatingRowDiagProduct, antiDiagLinear, antiDiagTensorEquiv, boundaryControl_DeficitVanishesAtDeltaEqD, charBandEval, cleanEntries, composeGridOps, computesWithin_toTransposeTM, coordShiftLinear, cutConeInputCells, cutTouchingOutputCells, cyclicBoundaryCount, cyclicChannelShift, encodeTMMatrixToGrid, evalConeTree, existsRecoverableSubsetBound, finsetNatDiam, fromAntiDiagLinear, gridSwapLinear, growingClassesPositiveDeficit, halfIntervalPartition4, leftComponentSubmodule, oneHopRecoverableInjective, orderedCubeMap, orderedRecoverableCells, outputBitIdx, rowShearLinear, rowShearTMMatrix, rowShearTransposeTM, shiftSetCovarianceSubmodule, shiftedFootprint, skewXorPad, stoneShuffleGridLinear, stoneShuffleTMMatrix, stoneShuffleTransposeTM, stoneUnshuffleGridLinear, stoneUnshuffleTMMatrix, stoneUnshuffleTransposeTM, theorem1_BoundedAritySetCapacity, theorem2BExplicitConst, theorem2BExplicitThreshold, threeTermBlockA, threeTermBlockB,

threeTermPaddedA, threeTermPaddedB, traceConeTree, treeCellSupport, trueConeSteps, uniformShannonMI, unrestrictedArity_ExactLogarithmicAsymptoticAllDimensions, unrestrictedArity_RateLowerBoundAndTendstoOne, walshBitSet, walshDigitMaskRotMatrix.

C Building and verifying

The repository must be cloned together with its sibling directories (`lean-machine` and the earlier papers of the series) so that relative Lake path dependencies resolve. Both `lean-machine` and `BoundedArityDecoders` pin Lean 4 and Mathlib to version `v4.34.1`.

To verify the shared shuffle, unshuffle, and row-shear machines in `../lean-machine`, run from the repository root:

```
cd lean-machine
lake exe cache get
lake build
lake env lean Axioms.lean
```

To regenerate the declaration index and axiom audit, verify all proofs in `lean/`, and build the manuscript PDF from this paper's directory, run:

```
python3 tools/lean_index.py
cd lean
lake exe cache get
lake build
lake env lean PaperAxioms.lean
cd ../paper
latexmk -pdf -interaction=nonstopmode -halt-on-error main.tex
```

Running `python3 tools/lean_index.py` regenerates `lean/BoundedArityDecoders.lean`, `lean/PaperAxioms.lean`, and Appendix B (`paper/sections/leanindex.tex`) directly from the `\formal` and `\lean` commands in `paper/sections/*.tex`. Every proved declaration audited in `Axioms.lean` and `PaperAxioms.lean` depends at most on Lean's three standard axioms `propext`, `Classical.choice`, and `Quot.sound`, with no `sorry`, `admit`, `native_decide`, `unsafe`, `implemented_by`, `extern`, or custom `axiom` declarations.

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